PROJECTIVE DECOMPOSITION AND INTERSECTION COHOMOLOGY OF RIGID SPACES

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Abstract. We prove the decomposition theorem for projective morphisms of rigid analytic spaces over p-adic fields, with $\mathbf{Q}_\ell$-coefficients for every prime ℓ, including $\ell = p$. The proof combines simultaneous algebraic approximation with a controlled restriction of perverse sheaves along analytic graphs, using vanishing cycles, Gysin duality, and comparison between coefficient primes. As an application, we show that the p-adic intersection cohomology of every proper rigid analytic space is de Rham. We also establish an independent criterion for de Rham intersection cohomology with local coefficients in terms of the dimension of the boundary, yielding further results for curves, surfaces, and finite covers.

Contents

1. Introduction 1
2. Approximation and boundary restriction 2
3. Projective decomposition 13
4. De Rham intersection cohomology 19
5. Examples and further remarks 25
References 29

1. Introduction

The decomposition theorem of Beilinson, Bernstein, and Deligne expresses the direct image of an intersection complex under a projective morphism of algebraic varieties as a direct sum of shifted intersection complexes [BBD82]. In rigid analytic geometry, Bhatt and Hansen developed the formalism of Zariski-constructible sheaves and perverse t-structures, and conjectured an analogous decomposition theorem for projective morphisms [BH22, Conjecture 4.17]. Their example of a proper smooth morphism from a nonarchimedean Hopf surface to the projective line shows that projectivity cannot be replaced by properness [BH22, §4.5].

We prove the rigid analytic analogue for every coefficient prime, including the residue characteristic. Let $K/\mathbf{Q}_p$ be a finite extension, $C = \widehat{\overline{K}}$, $G_K = \mathrm{Gal}(\overline{K}/K)$, and ℓ any prime. We use the rational constructible derived categories and middle perverse t-structures of [BH22], with $\mathcal{H}^q$ and ${}^p\mathcal{H}^a$ denoting ordinary and perverse cohomology sheaves. Write $X_C = X \times_K C$. Ambient spaces for rational coefficients are quasi-compact and quasi-separated (qcqs).

For a reduced pure-dimensional rigid space W, a smooth dense open immersion $j : U \hookrightarrow W$, and a finite-rank $\mathbf{Q}_\ell$-local system $\mathcal{L}$ for which $j_{!*}$ is defined, the intersection complex is $\mathrm{IC}_W(\mathcal{L}) = j_{!*}(\mathcal{L}[\dim W])$, where $j_{!*}$ denotes intermediate extension as in Section 2.1; put $\mathrm{IC}_W = \mathrm{IC}_W(\mathbf{Q}_\ell)$. For a smooth locally closed stratum $U \subset Y$, with reduced closure $\overline{U}$ and inclusions $j : U \hookrightarrow \overline{U}$ and $i : \overline{U} \hookrightarrow Y$, we write $\mathrm{IC}_{\overline{U}}(\mathcal{L}) = i_* j_{!*}(\mathcal{L}[\dim U])$. For non-equidimensional spaces, we take the direct sum of the intersection complexes of their irreducible components, pushed forward to the space.

Theorem 1.1. *Let $f : X \to Y$ be a projective morphism of rigid analytic spaces over K, with Y quasi-compact and quasi-separated. For every prime ℓ, including $\ell = p$, there is an isomorphism*

$$Rf_* \,\mathrm{IC}_X \simeq \bigoplus_a {}^p\mathcal{H}^a(Rf_* \,\mathrm{IC}_X)[-a], \tag{1.1}$$

and each perverse cohomology sheaf is a finite direct sum of intermediate extensions of finite-rank $\mathbf{Q}_\ell$-local systems on smooth Zariski-locally closed strata of Y.

This proves [BH22, Conjecture 4.17] in the stated setting. The local systems occurring here need not be semisimple. Indeed, we construct a smooth projective family of Tate elliptic curves over an annulus whose first cohomology local system remains nonsemisimple after extension to C.

The principal difficulty is to transfer the algebraic decomposition theorem to rigid analytic spaces while preserving perversity and the property of being an intermediate extension. Our approach begins with relative rigid GAGA and Popescu approximation, which locally realize a projective rigid morphism as the restriction of an algebraic projective morphism along an analytic section of a smooth family. The section can be expressed locally by successive hypersurface cuts. Although the algebraic decomposition theorem applies before restriction, restriction to an arbitrary hypersurface need not preserve perversity or intermediate extensions.

We simultaneously approximate the projective morphism, a prescribed closed boundary, and a resolution diagram of its inverse image. The approximation is arranged so that each successive cut is smooth on all the relevant smooth sources. For $\ell \neq p$, the results of Zhou on vanishing cycles [Zho25a], together with abstract blow-up triangles, show that the vanishing cycles of both the total direct image and its boundary restriction are zero. Uniform torsion estimates extend this vanishing to rational derived summands. Natural Gysin morphisms, constructed using the theory of cycle classes and duality developed by Li, Reinecke, and Zavyalov [LRZ24], provide the corresponding compatibility with Verdier duality.

The case $\ell = p$ requires a separate argument. We use the same algebraic approximation and analytic graph for all coefficient primes. Comparison between algebraic and Betti cohomology shows that the dimensions of ordinary stalk cohomology of the relevant algebraic perverse sheaves and their duals are independent of ℓ. The characterization of the perverse t-structure in terms of dimensions of supports transfers the upper perverse bounds to p, while Gysin duality supplies the lower bounds. This yields relative hard Lefschetz and the decomposition theorem for smooth sources. Resolution of singularities and stability under derived retracts then give the general result.

A first consequence concerns p-adic Hodge theory. With the perverse normalization, we write $\mathrm{IH}^m(X_C, \mathbf{Q}_p) = H^m(X_C, \mathrm{IC}_{X,\mathbf{Q}_p}|_{X_C})$.

Corollary 1.2. *For every proper rigid analytic space X/K and every integer m, the intersection cohomology group $\mathrm{IH}^m(X_C, \mathbf{Q}_p)$ is a de Rham representation of G_K.*

This establishes the assertion for proper spaces in [BH22, Conjecture 4.16(1)]. The proof realizes the intersection complex as a derived direct summand of the direct image of the constant sheaf under a projective resolution. The assertion then follows from p-adic Hodge comparison for smooth proper spaces.

We also obtain a de Rham result for local coefficients that is independent of the projective decomposition theorem. Let X/K be proper, reduced, and pure of dimension d, let $j : U \hookrightarrow X$ be a smooth dense Zariski-open immersion, and let $\mathbb{L}$ be a de Rham $\mathbf{Z}_p$-local system on U. Put $L = \mathbb{L}[1/p]$. We normalize intersection cohomology by $\mathrm{IH}^m(X_C, L) = H^m(X_C, (j_{!*}(L[d]))|_{X_C})$.

Theorem 4.3. *If $Z = X \setminus U$ is nonempty of dimension b, then $\mathrm{IH}^m(X_C, L)$ is de Rham whenever $m \leq -b$ or $m \geq b$. More precisely, the natural map from intersection cohomology to $H^{m+d}(U_C, L)$ is an isomorphism for $m < -b$ and injective for $m = -b$; the map from $H_c^{m+d}(U_C, L)$ to intersection cohomology is an isomorphism for $m > b$ and surjective for $m = b$.*

The proof combines logarithmic de Rham comparison on U with the strict support and cosupport conditions defining intermediate extension. In particular, a finite boundary gives the de Rham property in every degree. Normalization and finite covers yield further results for curves, surfaces, and local systems with finite geometric or divisorial monodromy.

The paper is organized as follows. Section 2 develops simultaneous algebraic approximation, boundary resolution, perverse restriction, and Gysin duality. Section 3 proves the projective decomposition theorem for all coefficient primes. Section 4 establishes the de Rham results, and Section 5 discusses examples and applications to singular support.

2. Approximation and boundary restriction

2.1. **Coefficients and splitting criteria.** For a coefficient ring Λ, write $D^b_{\mathrm{zc}}(M, \Lambda)$ for the bounded derived category of Zariski-constructible complexes of [BH22], and $\mathrm{Perv}(M, \Lambda)$ for its perverse heart. We write $\mathbb{D}_{M,\Lambda}$ for Verdier duality, omitting the space or coefficient ring when they are clear. Closures of supports are taken in the Zariski topology. For a qcqs space M, [BH22, Definition 4.8] defines the derived category of Zariski-constructible complexes with $\mathbf{Q}_\ell$-coefficients by $D^b_{\mathrm{zc}}(M, \mathbf{Q}_\ell) := D^b_{\mathrm{zc}}(M, \mathbf{Z}_\ell) \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell$. Writing $E_{\mathbf{Q}_\ell} = E[1/\ell]$, we

have, for $A, B \in D^b_{\mathrm{zc}}(M, \mathbf{Z}_\ell)$,

$$\mathrm{Hom}_{\mathbf{Q}_\ell}(A_{\mathbf{Q}_\ell}, B_{\mathbf{Q}_\ell}) \cong \mathrm{Hom}_{\mathbf{Z}_\ell}(A, B) \otimes_{\mathbf{Z}_\ell} \mathbf{Q}_\ell,$$

where the Hom groups are taken in the corresponding derived categories.

As in [BH22, Construction 4.3 and Proposition 4.6], there are two integral perverse t-structures, denoted by p and p^+, which are exchanged by Verdier duality. They induce the same rational perverse t-structure [BH22, Theorem 4.11]. For $n \geq 1$ and a complex A with $\mathbf{Z}_\ell$-coefficients, we write $A/\ell^n := A \otimes^L_{\mathbf{Z}_\ell} \mathbf{Z}/\ell^n$ for its derived reduction modulo ℓ^n. Since $\mathbf{Z}/\ell^n$ has Tor-amplitude contained in $[-1, 0]$ over $\mathbf{Z}_\ell$, this reduction sends $D^{[a,b]}$ into $D^{[a-1,b]}$ for ordinary cohomology.

Under the hypotheses of [BH22, Theorem 3.36], reduction modulo ℓ^n commutes with proper pushforward, ordinary pullback, extraordinary pullback along a closed immersion, and Verdier duality; see also [BH22, Remarks 3.33 and 3.37]. These operations also commute with extension of the base field. We use these compatibilities to work first with finite coefficients and then pass to $\mathbf{Z}_\ell$ and $\mathbf{Q}_\ell$. In particular, Li–Reinecke–Zavyalov [LRZ24, Theorem 7.5.18] prove that proper pushforward is compatible with Verdier duality for finite coefficients, including when $\ell = p$. They also show that, if W is smooth of pure dimension s, then its dualizing complex with $\mathbf{Z}/\ell^n$-coefficients is $(\mathbf{Z}/\ell^n)(s)[2s]$ [LRZ24, Corollary 7.1.2].

The trace and divisor class maps commute with reduction of coefficients. Their compatible systems modulo ℓ^n therefore define the corresponding ℓ-adic maps. By derived Nakayama [Stacks, Tag 0G1U], a morphism between derived ℓ-complete complexes is an isomorphism if its derived reduction modulo ℓ is an isomorphism; the integral isomorphism can then be rationalized. We apply this criterion to the purity map in Lemma 2.9 and to proper duality in Lemma 2.10.

Analytification is perverse t-exact for algebraic constructible complexes with finite coefficients [BH22, Theorem 4.2(4)]. For $\mathbf{Q}_\ell$-coefficients, we choose an integral algebraic model, analytify it, restrict it to the qcqs analytic open neighborhood under consideration, and then invert ℓ; this also applies when the full analytification is not quasi-compact. For an algebraic perverse object, we may first replace the model by its zeroth perverse cohomology with integral coefficients without changing its rationalization. Analytification with finite coefficients, followed by restriction to this analytic open, commutes with perverse truncation and Verdier duality. These compatibilities preserve the defining conditions for the integral p- and p^+-structures. After rationalization the two structures agree, so the resulting object is perverse [BH22, Construction 4.3 and proof of Theorem 4.11].

For a Zariski-open immersion $j : U \hookrightarrow M$ with M qcqs, we take intermediate extensions of extendable objects in the category of perverse sheaves with rational coefficients on M, as in [BH22, Theorems 4.2(5) and 4.11]. In particular, this applies to restrictions j^*P with $P \in \mathrm{Perv}(M, \mathbf{Q}_\ell)$.

Lemma 2.1. *Let A be an object in a derived category with a bounded t-structure preserved by the Tate twist (1), and write H^a for its cohomology functors. Let $\eta : A \to A(1)[2]$ be a morphism whose r-th power induces an isomorphism*

$$\eta^r : H^{-r}(A) \xrightarrow{\sim} H^r(A)(r) \qquad (r \geq 0).$$

Then $A \simeq \bigoplus_a H^a(A)[-a]$.

Proof. This is Deligne's splitting criterion [Del68, Del94]. Choose $n \geq 0$ such that $H^a(A) = 0$ for $a < -n$ and $a > n$. If $n = 0$, the assertion is immediate. Suppose $n > 0$, and write $B := H^{-n}(A)$ and $E := H^n(A)$. The bottom and top truncations give natural maps $u : B[n] \to A$ and $v : A \to E[-n]$, respectively. The hard Lefschetz isomorphism $\lambda := H^{-n}(\eta^n) : B \xrightarrow{\sim} E(n)$ allows us to construct a retraction $r : A \to B[n]$ and a section $s : E[-n] \to A$, with $ru = \mathrm{id}_{B[n]}$ and $vs = \mathrm{id}_{E[-n]}$. Define r as the composite

$$A \xrightarrow{\eta^n} A(n)[2n] \xrightarrow{v(n)[2n]} E(n)[n] \xrightarrow{\lambda^{-1}[n]} B[n].$$

To define s, take the composite

$$E(n)[n] \xrightarrow{\lambda^{-1}[n]} B[n] \xrightarrow{u} A \xrightarrow{\eta^n} A(n)[2n]$$

and apply the twist and shift $(-n)[-2n]$.

Since $n > 0$, we have $vu = 0$. Replacing s by $s' := s - urs$ gives $rs' = 0$ and $vs' = \mathrm{id}_{E[-n]}$. Thus $e_- := ur$ and $e_+ := s'v$ are idempotents with $e_-e_+ = e_+e_- = 0$. These orthogonal projectors split off the bottom and top cohomology objects simultaneously:

$$A \simeq B[n] \oplus A' \oplus E[-n],$$

where A' has cohomology only in degrees $-n+1, \ldots, n-1$.

Projecting η to A' gives a Lefschetz operator on A'. Its cohomology maps between the remaining degrees agree with those of η. For $a < n$, the powers from degree $-a$ to degree a involve only these remaining degrees, so the hard Lefschetz isomorphisms still hold. Applying the same argument to A' proves the assertion by induction. □

Lemma 2.2. *Let M be a qcqs rigid analytic space, let $P \in \mathrm{Perv}(M, \mathbf{Q}_\ell)$ be a perverse sheaf on M, and let $i : Z \hookrightarrow M$ be a closed immersion. Write $j : U := M \setminus Z \hookrightarrow M$ for the complementary open immersion. Define $B := {}^p\mathcal{H}^0(i^!P)$ and $C := {}^p\mathcal{H}^0(i^*P)$. The canonical morphisms $i_*B \xrightarrow{a} P \xrightarrow{b} i_*C$ define a morphism $c_Z(P) : B \to C$ by $b \circ a = i_* c_Z(P)$. Then $c_Z(P)$ is an isomorphism if and only if*

$$P \simeq j_{!*}j^*P \oplus i_*B. \tag{2.1}$$

When these conditions hold, the projector onto i_*B is $e := a \circ i_*\big(c_Z(P)^{-1}\big) \circ b$.

Proof. Since P is perverse, the t-exactness properties of closed immersions give $i^!P \in {}^pD^{\geq 0}(Z)$ and $i^*P \in {}^pD^{\leq 0}(Z)$. For every $Q \in \mathrm{Perv}(Z, \mathbf{Q}_\ell)$, adjunction and perverse truncation yield natural isomorphisms

$$\mathrm{Hom}_M(i_*Q, P) \simeq \mathrm{Hom}_Z(Q, i^!P) \simeq \mathrm{Hom}_Z(Q, B),$$
$$\mathrm{Hom}_M(P, i_*Q) \simeq \mathrm{Hom}_Z(i^*P, Q) \simeq \mathrm{Hom}_Z(C, Q).$$

Thus every morphism from a perverse sheaf supported on Z to P factors uniquely through a, and every morphism from P to a perverse sheaf supported on Z factors uniquely through b. The kernel of a and the cokernel of b are supported on Z, so these universal properties force both to vanish. Thus a is a monomorphism and b is an epimorphism. Consequently, i_*B is the maximal subobject of P supported on Z, while i_*C is its maximal quotient supported there.

Suppose first that $c_Z(P)$ is an isomorphism, and define $r := i_*\big(c_Z(P)^{-1}\big) \circ b : P \to i_*B$. Since $ba = i_*c_Z(P)$, we have $ra = \mathrm{id}_{i_*B}$. Hence a is a split monomorphism, and $e = ar$ is the projector onto i_*B. Setting $I := \ker r = \ker b$ gives $P \simeq i_*B \oplus I$. The summand I has no nonzero subobject supported on Z. Indeed, such a subobject would also be a subobject of P, hence would be contained in the maximal supported subobject i_*B, whereas $I \cap i_*B = 0$. To exclude quotients supported on Z, let $q : I \to i_*Q$ be any morphism with $Q \in \mathrm{Perv}(Z, \mathbf{Q}_\ell)$. Composing with the projection $P \to I$ gives a morphism $P \to i_*Q$. By the universal property of b, it factors through $b : P \to i_*C$. Since $b|_I = 0$, we have $q = 0$. Finally, $j^*i_*B = 0$, so the decomposition gives $j^*I \simeq j^*P$. Thus I extends j^*P and has neither nonzero subobjects nor nonzero quotients supported on Z. This characterizes the intermediate extension, giving $I \simeq j_{!*}j^*P$ and the decomposition (2.1).

Conversely, suppose that $P \simeq j_{!*}j^*P \oplus i_*B$. By the defining property of intermediate extension, $j_{!*}j^*P$ has no nonzero subobjects or quotients supported on Z. Hence ${}^p\mathcal{H}^0(i^!j_{!*}j^*P) = {}^p\mathcal{H}^0(i^*j_{!*}j^*P) = 0$. Full faithfulness of i_* gives $i^!i_*B \simeq B$ and $i^*i_*B \simeq B$. Under these identifications, the boundary morphism associated with i_*B is id_B. The boundary construction commutes with finite direct sums, so $c_Z(P)$ is an isomorphism.

□

Among perverse extensions of j^*P, the intermediate extension I is characterized by the strict bounds

$$i_*i^*I \in {}^pD^{\leq -1}(M), \qquad i_*i^*\mathbb{D}_M I \in {}^pD^{\leq -1}(M). \tag{2.2}$$

Indeed, the first condition excludes nonzero quotients supported on Z, while the second, by the self-duality of the perverse t-structure, excludes nonzero subobjects supported on Z. Both conditions use only restriction, perverse truncation, and Verdier duality on the ambient space.

Corollary 2.3. *Let M be a qcqs rigid space and let $P \in \mathrm{Perv}(M, \mathbf{Q}_\ell)$. Suppose that $c_Z(P)$ is an isomorphism for every Zariski-closed subspace $Z \subset M$. Then there are finitely many smooth Zariski-locally closed subspaces $U_\alpha \subset M$ and finite-rank $\mathbf{Q}_\ell$-local systems L_α on U_α such that*

$$P \simeq \bigoplus_\alpha \mathrm{IC}_{\overline{U}_\alpha}(L_\alpha).$$

Proof. We argue by induction on the dimension of the Zariski-closed support of P. The assertion is immediate when $P = 0$. Let $S = \overline{\mathrm{supp}(P)}$, with its reduced structure. Since closed pushforward identifies perverse sheaves on S with perverse sheaves on M supported on S, we may work on S. By constructibility, we can choose pairwise disjoint dense smooth open subsets of the irreducible components of S such that their union W is open in S and the restriction of P to each of them is a local system shifted by the dimension of the corresponding component. Let T be the closed complement of W in S.

By our assumption and the compatibility of the boundary construction with closed pushforward, the boundary morphism associated with T is an isomorphism. Writing $j_W : W \hookrightarrow S$, Lemma 2.2 gives $P \simeq j_{W!*}j_W^* P \oplus P'$, where P' is a perverse sheaf supported on T. Since W is a disjoint union of smooth open subsets, the first summand is a direct sum of intermediate extensions of shifted local systems.

It remains to treat the residual summand. For every closed subset of T, its boundary morphism is a direct summand of the corresponding boundary morphism for P, since the boundary construction commutes with finite direct sums. Consequently, these morphisms remain isomorphisms. Moreover, W is dense in every irreducible component of S, so $\dim \operatorname{supp}(P') \le \dim T < \dim S = \dim \operatorname{supp}(P)$. The induction hypothesis therefore applies to P'. Pushing the resulting decomposition forward from S to M gives the desired finite direct sum. □

2.2. **Simultaneous algebraic approximation.** We write $\mathrm{Spa}(B) = \mathrm{Spa}(B, B^\circ)$ for a K-affinoid algebra B, and S^{an} for analytification, taken relative to the corresponding affinoid space when S is projective over an affinoid algebra. For rigid spaces, $\Omega^1_{S/K}$ denotes continuous differentials.

Let $R = K[T_1, \dots, T_m]$, $A = K\langle T_1, \dots, T_m\rangle$, and $D = \mathrm{Spa}(A)$. We approximate finite diagrams of projective A-schemes by diagrams over a smooth algebraic variety. Here a finite diagram consists of finitely many projective A-schemes and A-morphisms between them. By [LRZ24, Lemma 7.1.4], the map $R \to A$ is regular, so Popescu's theorem [Stacks, Tag 07GC] expresses A as a filtered colimit of smooth R-algebras of finite type. The finitely many equations and morphisms defining the diagram descend to one such algebra.

Proposition 2.4. *Consider a finite diagram of projective A-schemes, some of whose vertices V_v have smooth rigid analytifications over K of pure dimensions d_v. There exist a smooth R-algebra B of finite type and constant relative dimension r, an R-algebra map $B \to A$, and an open subset $S \subset \operatorname{Spec} B$ containing the image of $\operatorname{Spec} A$, with the following properties:*

(i) *The entire finite diagram is defined over S. For each vertex V_v, write $V_{v,S}$ for its model over S, so that $V_{v,S} \times_S \operatorname{Spec} A \simeq V_v$.*

(ii) *If V_v is a vertex whose rigid analytification is smooth over K of pure dimension d_v, then its model $V_{v,S}$ is smooth over K of pure dimension $d_v + r$.*

(iii) *Put*

$$\mathcal{S} = S^{\mathrm{an}} \times_{(\mathbb{A}^m_K)^{\mathrm{an}}} D, \qquad \mathcal{V}_v = V_{v,S}^{\mathrm{an}} \times_{S^{\mathrm{an}}} \mathcal{S}.$$

The map $B \to A$ induces an analytic section $\gamma : D \to \mathcal{S}$. There are finitely many affinoid neighborhoods $N_\alpha \subset \mathcal{S}$ whose inverse images under γ cover D. On each neighborhood, write $N = N_\alpha$. There are functions $h_1, \dots, h_r$ on N that cut out $N \cap \gamma(D)$ as a closed analytic subspace. Write $V_N(h_1, \dots, h_k)$ for the closed analytic subspace defined by these functions, with $V_N(\varnothing) = N$, and put

$$\mathcal{V}_{v,k} = \mathcal{V}_v \times_{\mathcal{S}} V_N(h_1, \dots, h_k), \qquad 0 \le k \le r.$$

Every vertex and arrow of the diagram is restricted by base change to these same successive closed subspaces of N. Each $V_N(h_1, \dots, h_k)$ is smooth over K of pure dimension $m + r - k$. For $k < r$, the restriction of h_{k+1} to this space is smooth, so the next closed immersion is an effective Cartier divisor. Then $\mathcal{V}_{v,r}$ is the restriction of the rigid analytification of V_v to $\gamma^{-1}(N)$. For a vertex V_v as in (ii)*, each $\mathcal{V}_{v,k}$ is smooth over K of pure dimension $d_v + r - k$. For $k < r$, the restriction of h_{k+1} to $\mathcal{V}_{v,k}$ is smooth.*

We can also preserve any prescribed finite collection of projective embeddings, line bundles, closed immersions, Cartesian squares, nilpotent immersions, and isomorphisms over open complements. The analytic neighborhoods used for the cuts can be chosen qcqs.

Proof. Step 1. Descend the geometric data.

By relative projective GAGA [Con06, Example 3.2.6], projective rigid spaces over D come from projective A-schemes. The equations, morphisms, and identities in the finite diagram involve only finitely many coefficients in A. Popescu's theorem and descent for objects of finite presentation [Stacks, Tag 01ZM] give a smooth R-algebra B of finite type and an R-algebra map $B \to A$ such that all these data are defined over B. We retain the finitely presented ideals of the prescribed projective embeddings and closed immersions, the local trivializations and transition functions of the line bundles, and, for each nilpotent immersion with defining ideal $\mathcal{J}$, a relation $\mathcal{J}^n = 0$ with a fixed exponent n.

For each prescribed Cartesian square, we also descend the isomorphism from its upper-left vertex to the corresponding fibre product, together with its inverse and its compatibility with the two projections. For an isomorphism over an open complement, we first descend the defining ideal of the closed complement, and then the isomorphism and its inverse on the resulting open subschemes. These opens are quasi-compact because the

schemes are noetherian. After passing farther along the filtered system, all identities expressing inverses and compatibility hold simultaneously. Any further finite collection of data descends after passing farther along the same system.

Step 2. Make the descended sources smooth.

Fix a vertex V_v whose rigid analytification is smooth over K of pure dimension d_v, and choose an embedding $V_v \hookrightarrow \mathbb{P}^N_A$. At each closed point, relative analytification induces an isomorphism of completed local rings [Con06, Example 2.2.11]. Since the analytification of V_v is smooth, these completions are regular, so V_v is regular at its closed points. Excellence and the Jacobson property then imply that V_v is regular everywhere. At a closed point x, put $C_x = \mathcal{O}_{\mathbb{P}^N_A,x}$, and let J_x be the ideal of V_v. The local rings C_x and C_x/J_x are regular of dimensions $m+N$ and d_v, respectively. By [Stacks, Tag 00NR], J_x has $c_v = m+N-d_v$ generators that form part of a regular system of parameters of C_x. The comparison of completed local rings identifies their cotangent classes with cotangent vectors on the smooth analytic ambient space, so their continuous differentials are linearly independent at x. Choose finitely many affine open neighborhoods in $\mathbb{P}^N_A$ covering V_v on which these generators define V_v. On each neighborhood, let $\mathbf{q}$ be the column of defining generators retained in Step 1, and let $\mathbf{f} = (f_1, \dots, f_{c_v})^{\mathrm{t}}$ be the chosen generators. There are matrices P and Q of regular functions such that $\mathbf{f} = P\mathbf{q}$ and $\mathbf{q} = Q\mathbf{f}$. Descend these neighborhoods, functions, matrices, and both identities for all such vertices to one smooth R-algebra in the filtered presentation of A. The descended equations then generate the defining ideals on the descended neighborhoods. Since $\operatorname{Spec} A$ is connected, we may then replace B by an open-and-closed factor on which $\operatorname{Spec} B \to \operatorname{Spec} R$ has constant relative dimension r.

Write $V_{v,B}$ for the model of V_v over $\operatorname{Spec} B$. Let x be a closed point of V_v, and let x_B be its image in $V_{v,B}$. The residue field $\kappa(x)$ is finite over K, so x_B is also a closed point. On a chosen affine chart, pullback of functions followed by the continuous derivation defines a natural map from the algebraic cotangent space of $\mathbb{P}^N_B$ at x_B, after extension to $\kappa(x)$, to the continuous cotangent space of $(\mathbb{P}^N_A)^{\mathrm{an}}$ at x. It sends the differentials of the descended equations to those of the original equations. Their images are linearly independent, so the algebraic differentials are independent at x_B. Since the ambient $\mathbb{P}^N_B$ is smooth over K, the Jacobian criterion shows that $V_{v,B}$ is smooth at x_B, of dimension $(m+r+N)-c_v = d_v+r$.

Let $G_v \subset V_{v,B}$ be the union of the descended neighborhoods on which the chosen equations generate the defining ideal. Let $W_v \subset G_v$ be the open locus where $V_{v,B}$ is smooth over K of relative dimension d_v+r, and put $\mathcal{B}_v = V_{v,B} \setminus W_v$. Thus $\mathcal{B}_v$ is closed in the projective vertex $V_{v,B}$; it consists of the complement of G_v and the points of G_v where smoothness or the stated dimension fails. Its inverse image in V_v contains no closed points by the preceding argument. Since V_v is Jacobson, that inverse image is empty. Projectivity makes the image of $\mathcal{B}_v$ closed in $\operatorname{Spec} B$, and this image is disjoint from the image of $\operatorname{Spec} A$. Removing it for each vertex V_v whose rigid analytification is smooth of pure dimension d_v gives an open subset S such that $V_{v,S} = V_{v,B} \times_{\operatorname{Spec} B} S$ is smooth over K of pure dimension d_v+r.

Step 3. Express the original spaces as graph cuts.

The map $B \to A$ gives an analytic section $\gamma : D \to \mathcal{S}$ of the smooth projection $p : \mathcal{S} \to D$. Since $S \to \operatorname{Spec} R$ is smooth of relative dimension r, on affine open subsets it admits étale charts over $\operatorname{Spec} R$, with coordinates $s_1, \dots, s_r$ [Stacks, Tag 054L]. Analytification preserves separated and étale morphisms [Con99, Theorem 5.2.1(1)]; restricting these charts over D gives separated analytic étale charts near the section. Fix one such chart and an open subset $D_0 \subset D$ whose image under γ lies in that chart. Write $a_i(T) = \gamma^* s_i$ on D_0, and put $h_i = s_i - p^* a_i$ on the chart restricted to $p^{-1}(D_0)$, for $i = 1, \dots, r$. Their common zero locus is the pullback of the separated étale chart along the graph of $(a_1, \dots, a_r)$. Its projection to the corresponding open of D is therefore separated and étale, so the section is an open-and-closed subspace of this zero locus. Remove its complement in the zero locus from the chosen neighborhood, and denote the resulting neighborhood by N. Fix a vertex V_v as in (ii). The common zero locus of the pullbacks of $h_1, \dots, h_r$ to $\mathcal{V}_{v,0}$ is $\mathcal{V}_{v,r}$, which agrees with the restriction of the rigid analytification of V_v to $\gamma^{-1}(N)$ as a closed analytic subspace. Its ideal $\mathcal{I}$ is generated by these pullbacks.

At a classical point x of $\mathcal{V}_{v,r}$, the ambient analytic local ring and its quotient by $\mathcal{I}_x$ are regular of dimensions d_v+r and d_v. Let $\mathfrak{m}_x$ be the maximal ideal of the ambient analytic local ring. By the argument in Step 2 for regular local rings, $\mathcal{I}_x$ is generated by r members of a regular system of parameters. Since the pullbacks of $h_1, \dots, h_r$ also generate $\mathcal{I}_x$, their classes form a basis of $\mathcal{I}_x/\mathfrak{m}_x\mathcal{I}_x$ and have linearly independent images in $\mathfrak{m}_x/\mathfrak{m}_x^2$. Identifying this space with the continuous cotangent space shows that $dh_1, \dots, dh_r$ are independent at x. The closed analytic locus where these differentials are linearly dependent therefore has no classical point on $\mathcal{V}_{v,r}$, so its intersection with that fibre is empty.

Projectivity makes the image of this locus in the analytic base closed, and the preceding argument shows that it is disjoint from the section. After removing these images for the finitely many vertices V_v as in (ii), the map defined by $(h_1, \dots, h_r)$ is smooth on each $\mathcal{V}_{v,0}$. Thus $\mathcal{V}_{v,k}$ is smooth of pure dimension $d_v + r - k$. On this subspace, the remaining functions define a smooth map to $\mathbb{A}_K^{r-k,\mathrm{an}}$; for $k < r$, its composition with projection onto the first coordinate shows that h_{k+1} is smooth. The translated chart $(p, h_1, \dots, h_r) : N \to D_0 \times \mathbb{A}_K^{r,\mathrm{an}}$ is étale. Its restriction to $V_N(h_1, \dots, h_k)$ is étale over $D_0 \times \mathbb{A}_K^{r-k,\mathrm{an}}$, so this base is smooth of pure dimension $m + r - k$, and h_{k+1} is smooth on it for $k < r$. Thus each successive base cut is an effective Cartier divisor; its pullback to a vertex as in (ii) is the smooth Cartier divisor obtained above.

Step 4. Choose qcqs neighborhoods.

For every adic point of the section, choose an affinoid neighborhood in the analytic base on which the preceding construction is defined. The inverse images of these neighborhoods under γ cover D. Since D is quasi-compact, finitely many suffice; denote them by N_α. On each chosen affinoid base, the successive zero loci of the graph equations are closed affinoid subspaces. Restrict the entire finite diagram to these same bases, preserving its morphisms and identities at every stage. The corresponding sources remain projective over these bases and are therefore qcqs. □

For an arbitrary affinoid base Y, embed Y as a closed rigid subspace of a polydisc and compose the projective morphism with this embedding. A relatively ample line bundle H can be retained by descending an embedding associated with a sufficiently large power H^N. Since $c_1(H^N) = Nc_1(H)$, the a-th Lefschetz map is multiplied by the invertible scalar $N^a \in \mathbf{Q}_\ell^\times$, so the resulting map is an isomorphism with rational coefficients if and only if the original map is.

2.3. **Resolution of the boundary.** Repeated resolution and abstract blow-up triangles express the constant sheaf on a singular boundary in terms of proper direct images of constant sheaves on smooth spaces. Vanishing cycles commute with these proper direct images and triangles, so their vanishing on the smooth sources implies vanishing on the boundary. We include the resolution diagram in the algebraic approximation so that this argument applies after each graph cut.

Lemma 2.5. *Let V be a projective A-scheme. For any finite coefficient ring Λ, the constant sheaf Λ_V can be constructed from constant sheaves on finitely many regular projective A-schemes by proper pushforwards, shifts, and distinguished triangles. This construction is encoded by a finite diagram of projective A-schemes. The diagram and its defining identities descend to a common algebraic approximation and remain valid after every analytic graph cut.*

Proof. Step 1. Reduce to the case where V is reduced.

A nilpotent closed immersion induces an equivalence of étale topoi, identifying Λ_V with the pushforward of $\Lambda_{V_{\mathrm{red}}}$. Replace V by V_{red}, retaining the reduction immersion and a fixed nilpotence relation in the diagram so that this identification persists under base change. The resulting scheme is reduced, excellent, Noetherian, and finite-dimensional, of characteristic zero.

Step 2. Use resolution of singularities.

If V is regular, the assertion is immediate. Otherwise, Temkin's resolution theorem [Tem12, Theorem 1.2.1] gives a projective resolution $\pi : \widetilde{V} \to V$, obtained by a finite sequence of blow-ups whose centers are disjoint from the inverse image of the regular locus V_{reg}. Take Z to be the reduced closed complement of V_{reg}. Since V is reduced, this open locus contains the generic point of every irreducible component. Thus Z is nowhere dense and π is an isomorphism over $V \setminus Z$. The exceptional inverse image $E = \widetilde{V} \times_V Z$ gives a Cartesian square

$$\begin{array}{ccc} E & \longrightarrow & \widetilde{V} \\ \downarrow & & \downarrow \pi \\ Z & \xrightarrow{i} & V. \end{array}$$

Both Z and E have smaller dimension than V.

Step 3. Construct the abstract blow-up triangle.

For every finite coefficient ring Λ, this square gives a distinguished triangle

$$\Lambda_V \longrightarrow R\pi_*\Lambda_{\widetilde{V}} \oplus i_*\Lambda_Z \longrightarrow i_*R(\pi|_E)_*\Lambda_E \longrightarrow \Lambda_V[1]. \tag{2.3}$$

The first map consists of pullback to $\widetilde{V}$ and restriction to Z; the second is the difference of the two restrictions to E. Choose a bounded-below injective resolution $\Lambda_{\widetilde{V}} \to \mathcal{I}^\bullet$, put $\mathcal{B}^\bullet = \pi_*\mathcal{I}^\bullet$, and let $u : \Lambda_V \to \mathcal{B}^\bullet$ be the

adjunction map. Proper base change identifies $i_*i^*\mathcal{B}^\bullet$ with the third term of (2.3). Write $\mathrm{res} : \Lambda_V \to i_*\Lambda_Z$ and $\rho : \mathcal{B}^\bullet \to i_*i^*\mathcal{B}^\bullet$ for the restriction maps. In these representatives, the first two arrows are the maps of complexes

$$\alpha = (u, \mathrm{res}) : \Lambda_V \longrightarrow \mathcal{B}^\bullet \oplus i_*\Lambda_Z,$$
$$b = (\rho, -i_*i^*u) : \mathcal{B}^\bullet \oplus i_*\Lambda_Z \longrightarrow i_*i^*\mathcal{B}^\bullet,$$

Naturality gives $b\alpha = 0$ as a map of complexes. Hence b induces a global map $\beta : \mathrm{Cone}(\alpha) \to i_*i^*\mathcal{B}^\bullet$. Outside Z, the morphism π is an isomorphism, so α is a quasi-isomorphism and both sides of β are acyclic. At a geometric point $\bar{z}$ of Z, put $C_{\bar{z}} = R\Gamma(\widetilde{V}_{\bar{z}}, \Lambda)$. Proper base change identifies the stalk sequence with

$$\Lambda \longrightarrow C_{\bar{z}} \oplus \Lambda \longrightarrow C_{\bar{z}} \longrightarrow \Lambda[1],$$

where the arrows are (u, id) and $(a, b) \mapsto a - u(b)$, with $u : \Lambda \to C_{\bar{z}}$ the natural map. This is a split distinguished triangle, so β is a quasi-isomorphism at $\bar{z}$. Since geometric stalks detect quasi-isomorphisms, β is a global quasi-isomorphism. Identifying its target with $\mathrm{Cone}(\alpha)$ gives the distinguished triangle (2.3).

Step 4. Induction on dimension.

Apply the same procedure to Z_{red} and E_{red}. At each blow-up, the generic points of the irreducible components correspond bijectively to those of the preceding scheme [Stacks, Tag 0BFM], since the center misses all of them. The components of $\widetilde{V}$ are therefore the strict transforms of the components of V, with the same dimensions. The closed subsets $Z \subset V$ and $E \subset \widetilde{V}$ contain none of these generic points. Hence $\dim Z < \dim V$ and $\dim E < \dim \widetilde{V} = \dim V$. Induction on $\dim V$, the maximum dimension of its irreducible components, therefore terminates after finitely many abstract blow-up triangles. The terminal schemes are regular and projective over A; separate their finitely many open-and-closed pure-dimensional components. Relative analytification identifies completed local rings at classical points [Con06, Example 2.2.11], so their rigid analytifications are regular, hence smooth over the characteristic-zero field K. The triangles express Λ_V in terms of their constant sheaves by proper pushforwards and shifts.

Step 5. Preserve the construction after restriction to the graph.

The construction involves finitely many projective schemes, Cartesian squares, closed immersions, nilpotent immersions, and isomorphisms outside prescribed closed subsets. Proposition 2.4 descends this diagram and its relations to one smooth algebraic approximation. After each analytic graph cut, the squares remain Cartesian, the proper maps remain proper, and the maps remain isomorphisms outside the inverse images of the centers. The abstract blow-up triangles therefore remain valid. □

For a function $h : M \to \mathbb{A}^{1,\mathrm{an}}$ over a complete algebraically closed extension of K, put $M_0 = h^{-1}(0)$ and $M^\times = M \setminus M_0$, with inclusions $k : M_0 \hookrightarrow M$ and $j : M^\times \hookrightarrow M$. Let $q_e : M_e^\times \to M^\times$ be the pullback of $z \mapsto z^e$ on the punctured affine line, for $e \geq 1$. For a complex F with finite coefficients on $M^\times$, its nearby cycles are

$$\psi_h(F) = \varinjlim_e k^* R j_* R q_{e*} q_e^* F,$$

where the limit is indexed by divisibility. For a complex E on M with the same finite coefficients, its vanishing cycles $\phi_h(E)$ are defined by the distinguished triangle

$$k^*E \xrightarrow{\mathrm{sp}} \psi_h(j^*E) \longrightarrow \phi_h(E) \longrightarrow k^*E[1],$$

where sp is the specialization map induced by adjunction. Both functors carry the natural monodromy action of $\widehat{\mathbf{Z}}(1) = \varprojlim_e \mu_e$, where μ_e is the group of e-th roots of unity [Zho25a, Definition 2.1 and Remark 2.3].

Proposition 2.6. *Let $f : X \to D$ be projective, with X smooth over K and pure-dimensional, and let $i : Z \hookrightarrow D$ be a closed immersion. There is a simultaneous algebraic approximation as in Proposition 2.4 for which the following holds after extension to a completed algebraic closure of K, at every successive graph cut. Write $g : W \to M$ for the projective morphism at the current stage, $i_M : Z_M \hookrightarrow M$ for its boundary, and h for the next cut function. For every prime $\ell \neq p$ and $n \geq 1$, put $\Lambda_n = \mathbb{Z}/\ell^n$, $s = \dim W$, and $K_n = Rg_*\Lambda_{n,W}[s]$. Then*

$$\phi_h(K_n) = 0, \qquad \phi_h(i_{M*}i_M^*K_n) = 0. \tag{2.4}$$

The same approximation works for all primes $\ell \neq p$ and all n.

Proof. Step 1. Construct a single approximation containing the boundary.

Let $I \subset A$ be the finitely generated ideal defining Z, and put $Z_A = \mathrm{Spec}(A/I)$. By relative projective GAGA [Con06, Example 3.2.6], the morphism f is the rigid analytification of a projective morphism $f_A : X_A \to \mathrm{Spec}\, A$.

Set $(X_Z)_A = X_A \times_{\operatorname{Spec} A} Z_A$. Relative analytification commutes with fibre products [Con06, Example 2.2.11], so the analytification of $(X_Z)_A$ is $X_Z = X \times_D Z$.

Apply Lemma 2.5 to $(X_Z)_A$, and include its finite resolution diagram, f_A, the immersion $Z_A \hookrightarrow \operatorname{Spec} A$, and the Cartesian square defining $(X_Z)_A$ in one finite diagram. The vertex X_A and all regular terminal schemes have smooth rigid analytifications over K. Proposition 2.4 gives algebraic models on which each successive cut function is smooth on the model of X and every regular terminal source, after the preceding cuts. It also preserves the Cartesian squares and the relations defining the abstract blow-up squares.

Step 2. Vanishing cycles of the total direct image.

At a stage $g : W \to M$ of the successive cuts defining the graph, W is smooth over K, and the composite of g with the next cut function $h : M \to \mathbb{A}^{1,\mathrm{an}}_K$ is smooth on the chosen neighborhood. After passage to a completed algebraic closure, smooth local acyclicity [Zho25a, Example 2.14(1)] gives $\phi_{h\circ g}(\Lambda_{n,W}) = 0$ for $\ell \neq p$. Compatibility with proper pushforward [Zho25a, Lemma 2.12(2)] then gives

$$\phi_h(Rg_*\Lambda_{n,W}) \simeq Rg_{0*}\phi_{h\circ g}(\Lambda_{n,W}) = 0,$$

where g_0 is the induced proper map between the zero fibers. Shifting by $[s]$ gives $\phi_h(K_n) = 0$.

Step 3. Vanishing cycles of the boundary complex.

Put $W_Z = W \times_M Z_M$. Proper base change identifies

$$i_{M*}i_M^* Rg_*\Lambda_{n,W} \simeq R(g|_{W_Z})_*\Lambda_{n,W_Z},$$

where W_Z maps to M by its natural proper map. Use the finite resolution diagram of Lemma 2.5, whose abstract blow-up triangles remain valid after the preceding cuts. At a terminal vertex, the source and the next cut function on it are smooth by Proposition 2.4, so its constant sheaf has zero vanishing cycles. At an internal vertex, the abstract blow-up triangle expresses the constant sheaf in terms of those on a resolution, a center, and an exceptional inverse image, followed by proper pushforwards. Induction, compatibility with proper pushforward, and exactness of vanishing cycles give vanishing at that vertex. Proceeding back to the boundary gives $\phi_{(h\circ g)|_{W_Z}}(\Lambda_{n,W_Z}) = 0$. Compatibility with proper pushforward, followed by the shift $[s]$, gives $\phi_h(i_{M*}i_M^* K_n) = 0$, proving the second equality in (2.4).

Step 4. Repeat the argument after each graph cut.

The geometric relations included in the finite diagram remain valid after every successive graph cut, and the next cut function is smooth on every regular terminal source in the resolution diagram. Thus Steps 2 and 3 apply at every stage, for every $\ell \neq p$ and $n \geq 1$, with the same approximation. □

In the proof of Theorem 3.2, the two equalities in (2.4) preserve, respectively, perverse bounds after restriction to the graph and the strict boundary conditions characterizing intermediate extension.

2.4. **Perverse restriction and uniform torsion.** We work over a complete algebraically closed nonarchimedean field of characteristic zero and residue characteristic p, and fix a prime $\ell \neq p$. Let $h : M \to \mathbb{A}^{1,\mathrm{an}}$ be a morphism with M qcqs, let $k : M_0 \hookrightarrow M$ be its zero fiber, and put $T = k^*[-1]$. For finite coefficients prime to p, the shifted functors of nearby and vanishing cycles $\psi_h[-1]$ and $\phi_h[-1]$ are perverse t-exact on Zariski-constructible complexes of finite Tor-amplitude [Zho25a, Corollary 5.7 and Theorem 5.8(1)]. For nearby cycles, the complex on $M \setminus M_0$ must admit a constructible extension to M.

A complex $F \in D^b_{\mathrm{zc}}(M, \mathbb{Q}_\ell)$ has uniformly annihilated vanishing cycles along h if there exist an integral model $E \in D^b_{\mathrm{zc}}(M, \mathbb{Z}_\ell)$ of F and an integer $N \geq 0$ such that

$$\ell^N \operatorname{id}_{\phi_h(E \otimes^{\mathbf{L}}_{\mathbb{Z}_\ell} \mathbb{Z}/\ell^n)} = 0 \qquad \text{for every } n \geq 1. \tag{2.5}$$

We retain the monodromy action on nearby and vanishing cycles, and impose (2.5) in the corresponding derived category with finite coefficients [Zho25a, Remark 2.3(2)]. Perverse bounds refer to the underlying complexes on M_0 after forgetting monodromy.

Lemma 2.7. *Condition (2.5) is independent of the integral model of F. The class of rational complexes satisfying this condition is stable under finite direct sums, shifts, Tate twists, cones, and derived direct summands.*

Proof. Suppose that F' is a rational direct summand of F, and choose integral models E' and E, respectively. The inclusion and retraction have composite $\operatorname{id}_{F'}$. Since rational Hom groups are obtained by inverting ℓ in integral Hom groups, clearing denominators and multiplying by a further power of ℓ gives integral maps $u : E' \to E$ and $v : E \to E'$ with $vu = \ell^c \operatorname{id}_{E'}$ for a fixed integer $c \geq 0$. This is an equality in the integral derived category. If E satisfies (2.5) with exponent N, reduction modulo ℓ^n and application of vanishing cycles

give $\ell^{N+c}\,\mathrm{id}_{\phi_h(E'/\ell^n)} = 0$ for every $n \geq 1$. Thus F' satisfies (2.5). Applying this argument to an isomorphism between two rationalizations proves independence of the integral model. Finite direct sums, shifts, and Tate twists preserve the condition.

For cones, let $F_1 \to F_2 \to F_3 \to F_1[1]$ be a distinguished triangle of rational complexes, with F_1 and F_2 satisfying (2.5). Choose integral models E_1 and E_2. Multiplying the first morphism by a suitable power of ℓ gives an integral morphism $E_1 \to E_2$ whose cone is an integral model of F_3: multiplication by that power is invertible in the rational category and preserves the isomorphism class of the cone. Derived reduction and vanishing cycles give a distinguished triangle at each finite level. If the identities of the first two terms are annihilated by ℓ^u and ℓ^v, uniformly in n, the exact Hom sequences of the triangle show that the identity of the third term is annihilated by ℓ^{u+v}. This proves the assertion for cones. □

Lemma 2.8. *Suppose that $F \in D^b_{\mathrm{zc}}(M, \mathbb{Q}_\ell)$ satisfies (2.5). Then, for every integer a,*

$$F \in {}^pD^{\leq a} \quad \Longrightarrow \quad TF \in {}^pD^{\leq a},$$
$$F \in {}^pD^{\geq a} \quad \Longrightarrow \quad TF \in {}^pD^{\geq a}.$$

In particular, $k^*F[-1]$ is perverse whenever F is perverse and satisfies (2.5). This applies to rational perverse direct summands of shifts of integral complexes whose vanishing cycles are zero after reduction modulo every ℓ^n.

Proof. Step 1. Choose an integral model with the correct upper bound. Assume $F \in {}^pD^{\leq a}(M, \mathbb{Q}_\ell)$. Choose an integral model E_0 and put $E = {}^p\tau^{\leq a}E_0$. Rationalization is compatible with the perverse t-structure, so E still represents F, and Lemma 2.7 shows that it satisfies (2.5) for some exponent N. We have $E \in {}^pD^{\leq a}(M, \mathbb{Z}_\ell)$. By [BH22, Proposition 4.6(2)], each derived reduction $E_n = E \otimes^{\mathbf{L}}_{\mathbb{Z}_\ell} \mathbb{Z}/\ell^n$ belongs to ${}^pD^{\leq a}(M, \mathbb{Z}/\ell^n)$.

The reductions have finite Tor-amplitude over $\mathbb{Z}/\ell^n$, by [BH22, Remark 3.33]. Indeed,

$$E_n \otimes^{\mathbf{L}}_{\mathbb{Z}/\ell^n} \mathbb{F}_\ell \simeq E \otimes^{\mathbf{L}}_{\mathbb{Z}_\ell} \mathbb{F}_\ell$$

is bounded and Zariski-constructible, so E_n has finite Tor-amplitude, as required in [Zho25a, §5]. This also applies to integral models obtained by perverse truncation or Verdier duality, since both operations preserve the integral constructible category [BH22, Proposition 4.6(1) and Theorem 3.36(2)].

Step 2. Apply nearby and vanishing cycles. For a complex E on M, write $\Psi E = \psi_h(E|_{M\setminus M_0})[-1]$ and $\Phi E = \phi_h(E)[-1]$. By [Zho25a, Corollary 5.7 and Theorem 5.8(1)], both ΨE_n and ΦE_n belong to ${}^pD^{\leq a}(M_0, \mathbb{Z}/\ell^n)$; the assertion about nearby cycles applies because the complex on the open complement extends to E_n on M. The distinguished triangle

$$TE_n \longrightarrow \Psi E_n \longrightarrow \Phi E_n \longrightarrow TE_n[1]$$

gives $TE_n \in {}^pD^{\leq a+1}$, and its perverse long exact sequence identifies ${}^p\mathcal{H}^{a+1}(TE_n)$ as a quotient of ${}^p\mathcal{H}^a(\Phi E_n)$. Condition (2.5) annihilates ΦE_n by one power ℓ^N for all n, so $\ell^N\,{}^p\mathcal{H}^{a+1}(TE_n) = 0$ for every $n \geq 1$.

Step 3. Remove the possible extra perverse degree. Set $A = TE$. Ordinary pullback commutes with derived coefficient reduction, giving $A/\ell^n \simeq TE_n$. The bound for A/ℓ implies $A \in {}^pD^{\leq a+1}(M_0, \mathbb{Z}_\ell)$ by [BH22, Construction 4.3], and Step 2 gives $\ell^N\,{}^p\mathcal{H}^{a+1}(A/\ell^n) = 0$ for all n. Put $C = A[a]$ and $B = \mathbb{D}C$. Biduality identifies $\mathbb{D}B$ with C, so

$$\mathbb{D}B \in {}^pD^{\leq 1}, \qquad \ell^N\,{}^p\mathcal{H}^1((\mathbb{D}B)/\ell^n) = 0 \quad (n \geq 1).$$

These are the two conditions defining $B \in {}^pD^{\geq 0}$ in [BH22, Construction 4.3]. By duality, $C \in {}^{p^+}D^{\leq 0}$, hence $A \in {}^{p^+}D^{\leq a}$. Equivalently, ${}^p\mathcal{H}^{a+1}(A)$ is locally bounded torsion [BH22, Proposition 4.6(5)]. The integral p- and p^+-structures agree after inverting ℓ [BH22, Theorem 4.11], giving $TF = A[1/\ell] \in {}^pD^{\leq a}(M_0, \mathbb{Q}_\ell)$.

Step 4. Prove the lower perverse bound. Suppose $F \in {}^pD^{\geq a}(M, \mathbb{Q}_\ell)$. Self-duality of the perverse t-structure with rational coefficients gives $\mathbb{D}F \in {}^pD^{\leq -a}$. Choose an integral model $H \in {}^pD^{\leq -a}(M, \mathbb{Z}_\ell)$ of $\mathbb{D}F$ and put $E = \mathbb{D}H$, an integral model of F. By [BH22, Proposition 4.6(2)], $H_n = H/\ell^n$ belongs to ${}^pD^{\leq -a}$. Compatibility of Verdier duality with reduction of coefficients, together with perverse duality with finite coefficients, gives

$$E_n = E/\ell^n \simeq \mathbb{D}_{\mathbb{Z}/\ell^n}(H_n) \in {}^pD^{\geq a}.$$

Perverse t-exactness of Ψ and Φ with finite coefficients gives $\Psi E_n, \Phi E_n \in {}^pD^{\geq a}$. Since this subcategory is closed under fibers, the triangle in Step 2 yields $TE_n \in {}^pD^{\geq a}$.

Applying Verdier duality with finite coefficients and its compatibility with reduction, we obtain

$$(\mathbb{D}(TE))/\ell^n \simeq \mathbb{D}_{\mathbb{Z}/\ell^n}(TE_n) \in {}^pD^{\leq -a}.$$

Thus $\mathbb{D}(TE) \in {}^pD^{\leq -a}(M_0, \mathbb{Z}_\ell)$, so $TE \in {}^{p^+}D^{\geq a}$ by definition of the dual integral structure. Rationalization gives $TF \in {}^pD^{\geq a}(M_0, \mathbb{Q}_\ell)$. □

Remark. The lower bound holds for every $F \in {}^pD^{\geq a}(M, \mathbb{Q}_\ell)$, by perverse exactness with finite coefficients and Verdier duality. Condition (2.5) supplies the upper bound.

2.5. **Gysin maps and duality.** Let ℓ be any prime, including $\ell = p$, and let $k : M_0 \hookrightarrow M$ be an effective Cartier divisor. We work in the Zariski-constructible derived categories and assume that the ambient spaces are qcqs when using rational coefficients. For $n \geq 1$, write $\Lambda_n = \mathbb{Z}/\ell^n$. The localized divisor class of M_0 [LRZ24, Variant 3.1.5] gives a canonical morphism

$$\mathrm{cl}_{k,n} : k_*\Lambda_{n,M_0} \longrightarrow \Lambda_{n,M}(1)[2].$$

For a constructible Λ_n-complex F on M, tensoring with $F(-1)[-2]$ and applying the projection formula give a morphism $k_*(k^*F(-1)[-2]) \to F$. Its adjoint under $(k_*, k^!)$ defines the natural Gysin transformation

$$\tau_{k,F} : k^*F(-1)[-2] \longrightarrow k^!F. \tag{2.6}$$

We use the same notation for the integral and rational Gysin transformations constructed in the proof of Lemma 2.9.

Lemma 2.9. *Let $g : W \to M$ be projective, with W smooth over K of pure dimension s, and suppose that $W_0 = W \times_M M_0$ is a smooth effective Cartier divisor in W. Put $K_g = Rg_*\mathbb{Q}_\ell[s]$. Then τ_{k,K_g} is an isomorphism. The same holds for every derived direct summand of a shift or Tate twist of K_g, for every prime ℓ, including $\ell = p$.*

Proof. Step 1. Compare the Gysin maps under proper pushforward. Consider the Cartesian square

$$\begin{array}{ccc} W_0 & \xrightarrow{k'} & W \\ \downarrow g_0 & & \downarrow g \\ M_0 & \xrightarrow{k} & M. \end{array}$$

For finite coefficients Λ_n, ordinary and extraordinary proper base change give

$$k^*Rg_*\Lambda_{n,W} \simeq Rg_{0*}k'^*\Lambda_{n,W},$$
$$k^!Rg_*\Lambda_{n,W} \simeq Rg_{0*}k'^!\Lambda_{n,W}.$$

For a constructible Λ_n-complex F on W, [LRZ24, Lemma 7.3.4] gives the canonical equivalence $Rg_{0*}k'^!F \xrightarrow{\sim} k^!Rg_*F$. The second displayed isomorphism is its inverse for $F = \Lambda_{n,W}$. Indeed, closed base change gives $g^*k_* \simeq k'_*g_0^*$, and taking right adjoints gives this equivalence. Its adjoint under $(k_*, k^!)$ is the composite

$$k_*Rg_{0*}k'^!F \simeq Rg_*k'_*k'^!F \longrightarrow Rg_*F,$$

where the last arrow comes from the counit for k'. Under these identifications, the Gysin transformations fit into the commutative square

$$\begin{array}{ccc} k^*Rg_*\Lambda_{n,W}(-1)[-2] & \xrightarrow{\tau_{k,Rg_*\Lambda_{n,W}}} & k^!Rg_*\Lambda_{n,W} \\ \sim\downarrow & & \sim\downarrow \\ Rg_{0*}k'^*\Lambda_{n,W}(-1)[-2] & \xrightarrow{Rg_{0*}\tau_{k',\Lambda_{n,W}}} & Rg_{0*}k'^!\Lambda_{n,W}. \end{array} \tag{2.7}$$

Indeed, both k and k' are effective Cartier divisors, so $g^*\,\mathrm{cl}_{k,n} = \mathrm{cl}_{k',n}$ by [LRZ24, Lemma 3.1.6]. Taking the $(k_*, k^!)$-adjoints of the two composites in (2.7), and then their (g^*, Rg_*)-adjoints, identifies them through this equality and the projection formula.

Step 2. Prove purity on the smooth source. The spaces W and W_0 are smooth of dimensions s and $s-1$, respectively. Their dualizing complexes with Λ_n-coefficients are

$$\omega_W \simeq \Lambda_{n,W}(s)[2s], \qquad \omega_{W_0} \simeq \Lambda_{n,W_0}(s-1)[2s-2]$$

by [LRZ24, Corollary 7.1.2]. The trace map for k', applied to dualizing complexes, has adjoint $\omega_{W_0} \xrightarrow{\sim} k'^!\omega_W$. For k', the divisor class used above equals the cycle class for a local complete intersection of codimension one by [LRZ24, Lemma 3.3.6], applied to the single divisor W_0. After the twists and shifts in [LRZ24, Construction

7.2.5], the map defined by this cycle class agrees with the trace for k' on dualizing complexes by [LRZ24, Lemma 7.2.6]. Substituting these formulas and removing the common twist and shift gives

$$\tau_{k',\Lambda_{n,W}} : \Lambda_{n,W_0}(-1)[-2] \xrightarrow{\sim} k'^{!}\Lambda_{n,W}.$$

Applying Rg_{0*} and (2.7), we obtain that $\tau_{k,Rg_*\Lambda_{n,W}}$ is an isomorphism for every $n \geq 1$. The duality and trace results used here hold in mixed characteristic with $\mathbb{Z}/\ell^n$-coefficients for every prime ℓ, including $\ell = p$.

Step 3. Pass from finite coefficients to integral coefficients. To construct the integral divisor class, we first work in the stable ∞-category of complexes of abelian étale sheaves on M. Write $U = M \setminus M_0$ and $j : U \hookrightarrow M$, and let $\mathbb{G}_m$ denote the étale sheaf of invertible functions. The line bundle $\mathcal{O}_M(M_0)$, with its canonical trivialization on U, defines a Picard class with support in M_0 [LRZ24, Lemma 3.1.1]. At the level of mapping spaces, the trivialization is a nullhomotopy of the restriction of the line bundle to U; the cofiber sequence $j_!\mathbb{Z}_U \to \mathbb{Z}_M \to k_*\mathbb{Z}_{M_0}$ therefore gives a morphism $d_k : k_*\mathbb{Z}_{M_0} \to \mathbb{G}_{m,M}[1]$. Let $\mathcal{M}_\bullet = (\mu_{\ell^n})_n$ and $\mathcal{G}_\bullet = (\mathbb{G}_m)_n$, both with transition maps given by the ℓ-th power. For a sheaf F, let $\mathrm{const}(F)$ be the tower with value F and identity transition maps. The Kummer sequences form a short exact sequence of towers, in which $[\ell^n]$ is the ℓ^n-th power map at level n:

$$0 \longrightarrow \mathcal{M}_\bullet \longrightarrow \mathcal{G}_\bullet \xrightarrow{[\ell^n]} \mathrm{const}(\mathbb{G}_m) \longrightarrow 0.$$

The squares commute as maps of sheaves. Taking the connecting morphism in the derived ∞-category of towers and composing its shift with $\mathrm{const}(d_k)$ gives a morphism of towers $\mathrm{const}(k_*\mathbb{Z}_{M_0}) \to \mathcal{M}_\bullet[2]$.

Let $\mathcal{R}_\bullet = (\Lambda_n)_n$ be the tower of rings whose transition maps are the quotient maps. The tower $\mathcal{M}_\bullet$ is an $\mathcal{R}_\bullet$-module. The free-module adjunction in the stable ∞-category of towers turns the preceding morphism into an $\mathcal{R}_\bullet$-linear morphism

$$\mathrm{cl}_{k,\bullet} : (k_*\Lambda_{n,M_0})_n \longrightarrow (\mu_{\ell^n,M}[2])_n,$$

using $\Lambda_n \otimes^{\mathbf{L}}_{\mathbb{Z}} k_*\mathbb{Z}_{M_0} \simeq k_*\Lambda_{n,M_0}$. Both towers are derived-cartesian: derived extension of scalars from Λ_{n+1} to Λ_n identifies each term with the next one. For the target, this follows because $\mu_{\ell^{n+1}}$ is locally free of rank one over Λ_{n+1}, and the power map induces its reduction to μ_{ℓ^n}. The terms in both towers are bounded constructible complexes of finite Tor-dimension. The tower construction supplies the coherence required for a morphism in the inverse limit of categories in [BH22, Remark 3.33], and hence defines an integral class $\mathrm{cl}_{k,\mathbb{Z}_\ell} : k_*\mathbb{Z}_{\ell,M_0} \to \mathbb{Z}_{\ell,M}(1)[2]$. Its reduction modulo ℓ^n is the Kummer boundary of the line bundle with its trivialization on U, hence is $\mathrm{cl}_{k,n}$ by [LRZ24, Definition 3.1.4 and Variant 3.1.5].

For an integral constructible complex E, tensor this fixed class with $E(-1)[-2]$, apply the integral projection formula, and take the $(k_*, k^!)$-adjoint. This defines $\tau_{k,E} : k^*E(-1)[-2] \to k^!E$, naturally in every morphism of E. Put $E_n = E \otimes^{\mathbf{L}}_{\mathbb{Z}_\ell} \Lambda_n$. Compatibility with reduction of coefficients [BH22, Theorem 3.36(1), (3), (5)] identifies the reduction of $\tau_{k,E}$ with τ_{k,E_n}; part (5) applies to $k^!$ at every prime since k is finite. In particular, these reductions give the commutative square

$$\begin{array}{ccc} \big(k^*E_{n+1}(-1)[-2]\big) \otimes^{\mathbf{L}}_{\Lambda_{n+1}} \Lambda_n & \xrightarrow{\tau_{k,E_{n+1}} \otimes^{\mathbf{L}} 1} & (k^!E_{n+1}) \otimes^{\mathbf{L}}_{\Lambda_{n+1}} \Lambda_n \\ \sim\downarrow & & \sim\downarrow \\ k^*E_n(-1)[-2] & \xrightarrow{\tau_{k,E_n}} & k^!E_n. \end{array} \tag{2.8}$$

Take $E = Rg_*\mathbb{Z}_\ell[s]$. Proper pushforward commutes with coefficient reduction, so the reduction of $\tau_{k,E}$ modulo every ℓ^n is an isomorphism by Steps 1 and 2. Its cone is derived ℓ-complete and has zero reduction modulo every ℓ^n, so derived Nakayama makes $\tau_{k,E}$ an isomorphism. Inverting ℓ gives $\tau_{k,K_g} : k^*K_g(-1)[-2] \xrightarrow{\sim} k^!K_g$. The integral natural transformation also rationalizes: on the qcqs ambient space, rational morphisms are obtained from integral morphisms by inverting ℓ, so the resulting transformation is natural for all rational morphisms.

Step 4. Pass to derived direct summands. Let F be a derived direct summand of $K_g[a](t)$, with inclusion $u : F \to K_g[a](t)$ and retraction $v : K_g[a](t) \to F$ satisfying $vu = \mathrm{id}_F$. Naturality makes $\tau_{k,F}$ a retract of the isomorphism $\tau_{k,K_g[a](t)}$, and therefore an isomorphism. □

Lemma 2.10. *Let $f : X \to Y$ be a proper morphism of rigid analytic spaces over K, with Y qcqs. For every prime ℓ and every $E \in D^b_{\mathrm{zc}}(X, \mathbf{Z}_\ell)$, the proper duality maps with finite coefficients induce a natural isomorphism*

$$Rf_*\mathbb{D}_{X,\mathbf{Z}_\ell}E \xrightarrow{\sim} \mathbb{D}_{Y,\mathbf{Z}_\ell}(Rf_*E).$$

Its reduction modulo ℓ^n is the proper duality isomorphism with $\mathbf{Z}/\ell^n$-coefficients. Rationalization gives the corresponding natural isomorphism with $\mathbf{Q}_\ell$-coefficients.

Proof. Put $\Lambda_n = \mathbf{Z}/\ell^n$, and write $\omega_{X,n}$ and $\omega_{Y,n}$ for the dualizing complexes with Λ_n-coefficients, and $\omega_{X,\mathbf{Z}_\ell}$ and $\omega_{Y,\mathbf{Z}_\ell}$ for the integral dualizing complexes. The proof of [BH22, Theorem 3.36] gives identifications $\omega_{X,m} \otimes^{\mathbf{L}}_{\Lambda_m} \Lambda_n \simeq \omega_{X,n}$ for $m \geq n$, compatible with successive reductions, and likewise on Y. Proper pushforward commutes with these reductions.

We first check that the proper traces $\mathrm{Tr}_{f,n} : Rf_*\omega_{X,n} \to \omega_{Y,n}$ are compatible with reduction. The trace of a closed immersion i is the adjunction counit under the fixed identification $\omega_{X,n} \simeq i^!\omega_{Y,n}$, so it commutes with coefficient reduction. For smooth trace, reduction preserves the identities for composition and base change, the étale counit, and the algebraic trace on the projective line. The uniqueness in [LRZ24, Theorem 6.1.1] therefore gives compatibility for smooth trace. For a map with smooth source, the trace is the composite of the trace for the closed graph immersion and the trace for the smooth projection [LRZ24, Constructions 7.2.1, 7.2.7, and 7.3.1]. Consequently, reduction preserves all the normalizations, composition identities, and étale locality characterizing the proper traces in [LRZ24, Theorem 7.4.1]. Its uniqueness gives $\mathrm{Tr}_{f,m} \otimes^{\mathbf{L}}_{\Lambda_m} \Lambda_n = \mathrm{Tr}_{f,n}$.

By [LRZ24, Theorem 7.3.14], the internal Hom complex $R\mathcal{H}om(Rf_*\omega_{X,n}, \omega_{Y,n})$ lies in $D^{\geq 0}$. Applying derived global sections, which preserve this lower bound, gives $\mathrm{Ext}^{-q}(Rf_*\omega_{X,n}, \omega_{Y,n}) = 0$ for $q > 0$. Thus the mapping spaces containing these traces are discrete. The identifications under successive reductions therefore give unique homotopies realizing the reduction identities, and uniqueness makes these homotopies coherent. They define a morphism $\mathrm{Tr}_{f,\mathbf{Z}_\ell} : Rf_*\omega_{X,\mathbf{Z}_\ell} \to \omega_{Y,\mathbf{Z}_\ell}$ in the inverse limit of categories in [BH22, Remark 3.33]. Compose the natural evaluation map

$$Rf_*R\mathcal{H}om(E, \omega_{X,\mathbf{Z}_\ell}) \longrightarrow R\mathcal{H}om(Rf_*E, Rf_*\omega_{X,\mathbf{Z}_\ell})$$

with this trace. Compatibility of internal Hom and proper pushforward with coefficient reduction [BH22, Theorem 3.36(1), (3)] identifies its reduction with the duality map of [LRZ24, Definition 7.5.4], which is an isomorphism by [LRZ24, Theorem 7.5.18]. Its cone is derived ℓ-complete and has zero reduction modulo ℓ, so derived Nakayama proves the integral assertion. Localizing the integral functors and natural transformation at ℓ gives the rational assertion. □

Return to the setting and notation of Lemma 2.9. By Lemma 2.10 and smooth duality on W, we have $\mathbb{D}_M K_g \simeq K_g(s)$. Thus the class of shifted and Tate-twisted derived summands of K_g is preserved by Verdier duality. For such an object F, put $T = k^*[-1]$. Closed duality and Lemma 2.9 give

$$\begin{aligned} \mathbb{D}_{M_0}(TF) &\simeq k^!\mathbb{D}_M F[1] \\ &\simeq k^*\mathbb{D}_M F(-1)[-1] \\ &= T(\mathbb{D}_M F)(-1). \end{aligned} \tag{2.9}$$

For r successive transverse cuts defining the graph as in Proposition 2.4, let M_r be the final closed subspace and $\gamma : M_r \hookrightarrow M$ the composite immersion. Put $T^{(r)} = \gamma^*[-r]$. At each stage the source and its hypersurface cut are smooth, so Lemma 2.9 applies to the proper direct images and their derived summands. Iterating (2.9) gives

$$\mathbb{D}(T^{(r)}F) \simeq T^{(r)}(\mathbb{D}F)(-r). \tag{2.10}$$

We use these comparisons to check the strict cosupport conditions for intermediate extension after restriction to the analytic graph.

3. Projective decomposition

3.1. **Relative hard Lefschetz.** For a projective morphism with smooth source, we obtain relative hard Lefschetz by restricting the algebraic Lefschetz isomorphisms to the analytic graph supplied by Proposition 2.4. Lemma 2.8 shows that the perverse summands remain perverse after each successive hypersurface cut.

Theorem 3.1. *Let $f : X \to Y$ be a proper morphism with Y qcqs, equipped with a global relatively ample line bundle H on X, and assume that X is smooth over K of pure dimension d. Fix a prime $\ell \neq p$, put $K_f = Rf_*\mathbf{Q}_\ell[d]$, and let $\eta : K_f \to K_f(1)[2]$ be cup product with the first Chern class $c_1(H)$. Then the maps*

$$\eta^a : {}^p\mathcal{H}^{-a}(K_f) \xrightarrow{\sim} {}^p\mathcal{H}^a(K_f)(a) \qquad (a \geq 0) \tag{3.1}$$

are isomorphisms. Consequently, there is a derived decomposition $K_f \simeq \bigoplus_a {}^p\mathcal{H}^a(K_f)[-a]$ over K.

Proof. Step 1. Algebraic approximation. The assertion that the canonical Lefschetz maps are isomorphisms is local on Y. Restrict to an affinoid open subset, over which a sufficiently high power of H gives a projective embedding, and embed the affinoid as a closed subspace of a polydisc D. Closed pushforward is fully faithful

and perverse t-exact, so we may prove the assertion for the composite map to D. Apply Proposition 2.4, retaining the projective embedding and a positive tensor power $H^{\otimes N}$. We obtain a smooth algebraic K-variety S, a projective morphism $g_0 : W_0 \to S$, and local cuts defining an analytic graph that recovers the original morphism. If the graph has codimension r, then W_0 is smooth of pure dimension $s = d + r$. On each qcqs neighborhood supplied by the proposition, every intermediate source is smooth, and the next cut function is smooth on that source.

Step 2. Algebraic hard Lefschetz. Put $K_0 = Rg_{0*}\mathbf{Q}_\ell[s]$ and $P_{a,0} = {}^p\mathcal{H}^a(K_0)$. Let η_0 be cup product with the first Chern class of the retained relatively ample line bundle on W_0. The algebraic relative hard Lefschetz theorem [BBD82, Theorem 6.2.10] gives $\eta_0^a : P_{-a,0} \xrightarrow{\sim} P_{a,0}(a)$. These morphisms are defined over K, and their geometric base changes are isomorphisms, so they are isomorphisms over K. Lemma 2.1 gives a derived decomposition

$$K_0 \simeq \bigoplus_a P_{a,0}[-a].$$

In particular, each $P_{a,0}[-a]$ is a direct summand of K_0 in the rational derived category. On a chosen qcqs analytic neighborhood of the graph, form the restricted analytifications from integral models as in Section 2.1, and then invert ℓ. For each algebraic perverse object, replace an integral model by its integral perverse cohomology in degree zero; this has the same rationalization. Analytification with finite coefficients is perverse t-exact by [BH22, Theorem 4.2(4)]. Construction 4.3 and the proof of Theorem 4.11 of [BH22] then show that the object obtained by restriction and rationalization is perverse. Compatibility with proper pushforward [BH22, Proposition 3.16] identifies the ambient complex with the analytic direct image, and the functor preserves the displayed decomposition. We use the same notation after restriction.

Step 3. Restriction to the graph. Pass to a completed algebraic closure to apply the results on vanishing cycles with finite coefficients. Write the bases obtained after successive cuts as $M_0 \supset M_1 \supset \cdots \supset M_r$, with M_0 the chosen analytic neighborhood. For $0 \le j < r$, let h_j define the next cut $k_j : M_{j+1} \hookrightarrow M_j$, and put $T_j = k_j^*[-1]$. For $0 \le j \le r$, let $W_j = W_0 \times_{M_0} M_j$ and $g_j : W_j \to M_j$ be the corresponding base changes, and put $K_j = Rg_{j*}\mathbf{Q}_\ell[s-j]$ and $E_j = Rg_{j*}\mathbf{Z}_\ell[s-j]$. Proposition 2.4 makes each W_j smooth of pure dimension $s-j$, and makes $h_j \circ g_j$ smooth for $j < r$. For $j < r$, proper base change identifies $T_j K_j \simeq K_{j+1}$.

We prove by induction that

$$K_j \simeq \bigoplus_a P_{a,j}[-a], \qquad P_{a,j} = T_{j-1}\cdots T_0 P_{a,0} \in \mathrm{Perv}(M_j, \mathbf{Q}_\ell),$$

with the empty composite understood as the identity. The case $j = 0$ follows from Step 2. Suppose the assertion holds at stage j. For $\Lambda_n = \mathbf{Z}/\ell^n$, coefficient reduction gives $K_{j,n} := E_j \otimes^{\mathbf{L}}_{\mathbf{Z}_\ell} \Lambda_n \simeq Rg_{j*}\Lambda_{n,W_j}[s-j]$. Smooth local acyclicity and compatibility with proper pushforward give $\phi_{h_j}(K_{j,n}) = 0$ for every n, as in Proposition 2.6. Each $P_{a,j}$ is a shifted rational derived summand of K_j, so Lemma 2.7 gives condition (2.5) for $P_{a,j}$. Lemma 2.8 then shows that $P_{a,j+1} := T_j P_{a,j}$ is perverse. Applying T_j to the decomposition of K_j gives $K_{j+1} \simeq \bigoplus_a P_{a,j+1}[-a]$, completing the induction. At the final graph, write $\gamma : M_r \hookrightarrow M_0$ for the composite of the cuts and put $T^{(r)} = T_{r-1}\cdots T_0 = \gamma^*[-r]$. On the part of the original base covered by this neighborhood, we obtain

$$\begin{aligned} K_f &\simeq T^{(r)}K_0 \simeq \bigoplus_a T^{(r)}P_{a,0}[-a], \\ {}^p\mathcal{H}^a(K_f) &\simeq T^{(r)}P_{a,0}. \end{aligned} \tag{3.2}$$

Step 4. The Lefschetz maps after restriction. The retained algebraic line bundle restricts to $H^{\otimes N}$. Since first Chern classes commute with pullback, its cup-product operator restricts to $N\eta$. To identify the induced maps on perverse cohomology, choose the decomposition of K_0 in Step 2 to induce the identity on its perverse cohomology objects, and express $\eta_0^a : K_0 \to K_0(a)[2a]$ as a matrix relative to this decomposition. Its component from perverse degree $-a$ to degree a is the algebraic Lefschetz isomorphism $P_{-a,0} \to P_{a,0}(a)$. Every transported summand remains perverse, so applying $T^{(r)}$ to this component gives the induced map on ${}^p\mathcal{H}^{-a}(K_f)$ and ${}^p\mathcal{H}^a(K_f)(a)$. This map is $N^a\eta^a$ under (3.2) and is an isomorphism. Since N^a is invertible in $\mathbf{Q}_\ell$, equation (3.1) follows.

Step 5. Descent and global splitting. Write $q_Y : Y_C \to Y$ and $q_X : X_C \to X$ for extension to the completed algebraic closure. Proper pushforward and Verdier duality commute with this base change [BH22, Proposition 3.24 and Theorem 3.36]. Base extension preserves the dimensions of constructible supports and detects vanishing on geometric stalks. On the qcqs ambient spaces, finite constructible stratifications give the same detection after rationalization. The support criterion and duality therefore make q_Y^* conservative and perverse t-exact [BH22,

Definition 4.1, Construction 4.3, and Theorem 4.11]. Naturality of the Kummer connecting morphisms gives $q_X^* c_1(H) = c_1(q_X^* H)$, and proper base change identifies the associated cup-product operators. Thus the Lefschetz maps over C are the pullbacks of the canonical maps over K, whose invertibility follows by conservativity. They are defined globally by $c_1(H)$, so local invertibility proves (3.1) on Y. Applying Lemma 2.1 in the global rational derived category with its perverse t-structure gives $Rf_*\mathbf{Q}_\ell[d] \simeq \bigoplus_a {}^p\mathcal{H}^a(Rf_*\mathbf{Q}_\ell[d])[-a]$. □

3.2. **Splitting along the boundary.** The algebraic decomposition by supports passes through the cuts defining the graph once the summands given by intermediate extension retain their strict bounds on the boundary, both before and after Verdier duality.

Theorem 3.2. *Under the assumptions of Theorem 3.1, let $P_a = {}^p\mathcal{H}^a(Rf_*\mathbf{Q}_\ell[d])$, with $\ell \neq p$. For every closed immersion $i : Z \hookrightarrow Y$, the canonical boundary map*

$$c_Z(P_a) : {}^p\mathcal{H}^0(i^! P_a) \longrightarrow {}^p\mathcal{H}^0(i^* P_a)$$

is an isomorphism. Consequently, each P_a is a finite direct sum of intermediate extensions of finite-rank local systems on smooth Zariski-locally closed subspaces of Y.

Proof. Step 1. Approximation with a prescribed boundary. Fix $i : Z \hookrightarrow Y$. The assertion is local on Y, so restrict to an affinoid open subset, embed it in a polydisc, and apply Proposition 2.6. This gives a projective algebraic morphism $g_0 : W_0 \to S$, a closed boundary $i_0 : Z_0 \hookrightarrow S$, and a finite resolution diagram for its inverse image in W_0, all in one approximation. The proposition supplies both vanishing statements in (2.4) at every successive cut of this diagram. Let r be the codimension of the analytic graph and put $s = \dim W_0 = d + r$. Write $K_0 = Rg_{0*}\mathbf{Q}_\ell[s]$, $P_{a,0} = {}^p\mathcal{H}^a(K_0)$, and let j_0 be the open complement of i_0.

Step 2. Algebraic boundary splitting. The algebraic decomposition theorem [BBD82, Theorem 6.2.5] makes $P_{a,0}$ geometrically semisimple. Over an algebraic closure, each simple constituent is either supported on Z_0 or is the intermediate extension of its restriction to $S \setminus Z_0$. Hence the canonical boundary map $c_{Z_0}(P_{a,0})$ becomes an isomorphism after geometric base change. It is defined over K, so it is an isomorphism over K. Lemma 2.2 gives the canonical decomposition

$$P_{a,0} \simeq I_{a,0} \oplus i_{0*}Q_{a,0}, \qquad I_{a,0} = j_{0!*}j_0^* P_{a,0}, \tag{3.3}$$

where $Q_{a,0}$ is perverse on Z_0. The summand $I_{a,0}$ satisfies

$$i_{0*}i_0^* I_{a,0} \in {}^pD^{\leq -1}, \qquad i_{0*}i_0^* \mathbb{D} I_{a,0} \in {}^pD^{\leq -1}.$$

By (2.2), these conditions express the absence of nonzero quotients and subobjects supported on Z_0. The restricted analytification of Section 2.1 preserves the decomposition and these bounds.

Step 3. Induction through the cuts. On a chosen qcqs analytic neighborhood, pass to a completed algebraic closure and use $M_j, W_j, g_j, h_j, k_j, K_j, K_{j,n}, T_j$, and $P_{a,j}$ as in the proof of Theorem 3.1. Let $i_j : Z_j \hookrightarrow M_j$ be the boundary at stage j, with complementary open immersion $j_j : M_j \setminus Z_j \hookrightarrow M_j$. We prove inductively that

$$P_{a,j} \simeq I_{a,j} \oplus i_{j*}Q_{a,j}, \qquad I_{a,j} \in \mathrm{Perv}(M_j, \mathbf{Q}_\ell), \quad Q_{a,j} \in \mathrm{Perv}(Z_j, \mathbf{Q}_\ell),$$
$$i_{j*}i_j^* I_{a,j} \in {}^pD^{\leq -1}, \qquad i_{j*}i_j^* \mathbb{D} I_{a,j} \in {}^pD^{\leq -1}.$$

We also retain that $I_{a,j}[-a]$ and $i_{j*}Q_{a,j}[-a]$ are derived retracts of K_j. These assertions hold at $j = 0$ by Step 2 and the algebraic derived decomposition.

Step 4. Perversity and the first boundary bound. Assume the assertions at stage j, and consider $T_j = k_j^*[-1]$. Proposition 2.6 gives $\phi_{h_j}(K_{j,n}) = 0$ for every n. The objects $I_{a,j}$ and $i_{j*}Q_{a,j}$ are perverse and are shifted rational retracts of K_j, so Lemmas 2.7 and 2.8 show that $I_{a,j+1} := T_j I_{a,j}$ and $T_j(i_{j*}Q_{a,j})$ are perverse. Closed base change shows that the latter is supported on Z_{j+1}.

Put $B_j = i_{j*}i_j^* K_j$. The second vanishing statement in Proposition 2.6 gives $\phi_{h_j}(i_{j*}i_j^* K_{j,n}) = 0$ for every n. The complex $i_{j*}i_j^* I_{a,j}$ is a shifted rational retract of B_j, and the induction hypothesis places it in ${}^pD^{\leq -1}$. Lemmas 2.7 and 2.8 preserve this upper bound under T_j. Closed base change therefore gives

$$i_{j+1*}i_{j+1}^* I_{a,j+1} \simeq T_j(i_{j*}i_j^* I_{a,j}) \in {}^pD^{\leq -1}. \tag{3.4}$$

Step 5. The dual boundary bound. Smoothness of W_j and compatibility of proper pushforward with Verdier duality give $\mathbb{D}K_j \simeq K_j(s-j)$. Thus $\mathbb{D}I_{a,j}$ is a shifted and Tate-twisted rational retract of K_j, and its boundary restriction is a corresponding retract of B_j. The induction hypothesis gives $i_{j*}i_j^* \mathbb{D} I_{a,j} \in {}^pD^{\leq -1}$. Applying the boundary vanishing argument of Step 4 yields $T_j(i_{j*}i_j^* \mathbb{D} I_{a,j}) \in {}^pD^{\leq -1}$. The next graph cut is a Cartier divisor

whose pullback to W_j is a smooth Cartier divisor, so Lemma 2.9 and (2.9) apply to these retracts. They give $\mathbb{D}(T_j I_{a,j}) \simeq T_j(\mathbb{D}I_{a,j})(-1)$. Combining this with closed base change, we obtain

$$i_{j+1*}i_{j+1}^*\mathbb{D}I_{a,j+1} \simeq T_j(i_{j*}i_j^*\mathbb{D}I_{a,j})(-1) \in {}^pD^{\leq -1},$$

since Tate twists preserve the perverse t-structure.

Step 6. Intermediate extension after restriction. The perverse sheaf $I_{a,j+1}$ satisfies both strict boundary conditions, hence $I_{a,j+1} \simeq j_{j+1!*}j_{j+1}^*I_{a,j+1}$. Applying T_j to the decomposition at stage j gives

$$P_{a,j+1} \simeq I_{a,j+1} \oplus T_j(i_{j*}Q_{a,j}).$$

Both summands are perverse, and the second is supported on Z_{j+1}. Closed pushforward therefore identifies it with $i_{j+1*}Q_{a,j+1}$ for a perverse sheaf $Q_{a,j+1}$ on Z_{j+1}. Lemma 2.2 makes $c_{Z_{j+1}}(P_{a,j+1})$ an isomorphism. The shifted summands remain retracts of $T_jK_j \simeq K_{j+1}$, completing the induction.

Step 7. Descent to the rigid base. At the final graph cut, (3.2) identifies $P_{a,r}$ with ${}^p\mathcal{H}^a(K_f)$. The induction shows that its canonical boundary map for Z is an isomorphism. Since closed extraordinary pullback and duality commute with extension to the completed algebraic closure, this assertion descends to K. Invertibility can be checked locally, so $c_Z(P_a)$ is an isomorphism on Y. Applying the construction for each closed Z and then Corollary 2.3 expresses P_a as a finite direct sum of intermediate extensions of finite-rank local systems on smooth Zariski-locally closed strata. Together with the global derived splitting of Theorem 3.1, this gives the projective decomposition for smooth sources and $\ell \neq p$. □

3.3. **Coefficients in the residue characteristic.** To extend the decomposition for smooth sources proved in Sections 3.1 and 3.2 to $\mathbf{Q}_p$-coefficients, we keep the algebraic approximation and analytic graph fixed for all coefficient primes. The algebraic complexes arise from a single Betti complex with rational coefficients, so their ordinary stalk dimensions are independent of the prime. The support criterion transfers to p the upper perverse bounds obtained after restriction to the graph; applying the Gysin comparison of Section 2.5 to the algebraic duals gives the lower bounds.

Lemma 3.3. *Let $g_0 : W_0 \to S$ be a projective morphism of K-schemes of finite type, with W_0 smooth of pure dimension s. Fix a closed immersion $i_0 : Z_0 \hookrightarrow S$, with complementary open immersion $j_0 : S \setminus Z_0 \hookrightarrow S$. For every prime ℓ, put $K_{0,\ell} = Rg_{0*}\mathbf{Q}_\ell[s]$, $P_{a,\ell} = {}^p\mathcal{H}^a(K_{0,\ell})$, and $I_{a,\ell} = j_{0!*}j_0^*P_{a,\ell}$. At corresponding geometric closed points of S, the dimensions of ordinary stalk cohomology are independent of ℓ for*

$$P_{a,\ell}, \quad \mathbb{D}P_{a,\ell}, \quad I_{a,\ell}, \quad \mathbb{D}I_{a,\ell}, \quad i_{0*}i_0^*I_{a,\ell}, \quad i_{0*}i_0^*\mathbb{D}I_{a,\ell}. \tag{3.5}$$

Proof. Choose an abstract embedding $\overline{K} \hookrightarrow \mathbf{C}$, where $\mathbf{C}$ is the field of complex numbers. Such an embedding exists since $\overline{K}$ has characteristic zero and cardinality at most that of $\mathbf{C}$. Write $g_{0,\mathbf{C}} : W_{0,\mathbf{C}} \to S_{\mathbf{C}}$ for the base change along this embedding, and consider the Betti complex $K_B = R(g_{0,\mathbf{C}})_*\mathbf{Q}[s]$ on the underlying complex analytic space of $S_{\mathbf{C}}$. Take its perverse cohomology, Verdier duals, intermediate extensions, and restrictions to the prescribed boundary in the constructible derived category on the complex variety $S_{\mathbf{C}}$. These constructions are defined over $\mathbf{Q}$ and commute with extension of the coefficient field. In particular, intermediate extension is the image of the natural morphism between the perverse extensions, and formation of this image commutes with coefficient extension.

The comparison between algebraic and classical cohomology with finite coefficients [SGA4, Exposé XVI, Theorem 4.1] extends to constructible $\mathbf{Z}_\ell$- and $\mathbf{Q}_\ell$-complexes, compatibly with the cohomological operations, by [BBD82, §6.1.2, (B'), (B"), and (C')]. The support and cosupport conditions make this comparison perverse t-exact, so it commutes with perverse cohomology and intermediate extension. Moreover, the boundary functors i_0^* and i_{0*} commute with flat extension of the Betti coefficient field from $\mathbf{Q}$ to $\mathbf{Q}_\ell$. These compatibilities identify all the $\mathbf{Q}_\ell$-realizations in (3.5) with the corresponding geometric algebraic complexes. They apply to every prime, including p, since the varieties have characteristic zero. Each ordinary stalk cohomology group is therefore obtained from a finite-dimensional rational Betti group by extension of scalars, proving the assertion. □

Lemma 3.4. *Let $V = \mathrm{Spa}(A_V)$ be an affinoid rigid space over K, and put $\mathcal{V} = \operatorname{Spec} A_V$. Supports and their closures are taken in $\mathcal{V}$, using affinoid comparison for objects on V. Choose finitely many coefficient primes and finitely many integral constructible complexes G_ℓ on $\mathcal{V}$ for each prime. Write $F_\ell = G_\ell[1/\ell]$, also using these symbols for the corresponding objects on V, and suppress any additional index for complexes at the same prime. There is a common finite stratification $\mathcal{V} = \coprod_b S_b$ by connected regular affine locally closed*

subschemes such that every $G_\ell|_{S_b}$ is pro-étale locally constant with a fixed perfect $\mathbf{Z}_\ell$-complex $C_{\ell,b}$ as its value. Put $e_b = \dim S_b = \dim \overline{S_b}$ and $M^q_{\ell,b} = H^q(C_{\ell,b})$. For every integer a, the following conditions are equivalent:

$$F_\ell \in {}^pD^{\leq a}(V, \mathbf{Q}_\ell); \tag{3.6}$$

$$M^q_{\ell,b}[1/\ell] = 0 \quad \text{whenever } q > a - e_b; \tag{3.7}$$

$$\dim \overline{\operatorname{supp} \mathcal{H}^q(F_\ell)} \leq a - q \quad \text{for every } q. \tag{3.8}$$

Proof. Integral affinoid comparison [BH22, Lemma 3.34] allows us to work on $\mathcal{V}$. By [BS14, Proposition 6.6.11], the finitely many complexes admit finite constructible stratifications on which they are pro-étale locally constant and perfect. Take a common refinement, and use excellence and finite dimension of A_V to refine it into connected regular affine strata compatible with a finite filtration by closed subschemes. On each stratum, [BS14, Lemma 6.6.8] gives a fixed perfect value $C_{\ell,b}$. A connected regular stratum is irreducible, and in this Jacobson setting its dimension equals the dimension of its closure.

For finite coefficients, [BH22, Definition 4.1], together with Construction 4.3 and Example 4.4 for integral coefficients, identifies the upper perverse bound for a locally constant complex on a regular stratum of dimension e_b with the ordinary bound $H^q = 0$ for $q > a - e_b$. These bounds glue along the chosen finite filtration by closed subschemes as in [BH22, proof of Proposition 4.6(1)].

Suppose (3.7) holds. Fix ℓ, and put $G_b = G_\ell|_{S_b}$ and $c_b = a - e_b$. Consider the ordinary truncation triangle

$$\tau^{\leq c_b} G_b \longrightarrow G_b \longrightarrow \tau^{>c_b} G_b \longrightarrow (\tau^{\leq c_b} G_b)[1].$$

The complex $\tau^{>c_b} G_b$ has perfect value $\tau^{>c_b} C_{\ell,b}$. Its cohomology modules are finitely generated and become zero after rationalization, so one power ℓ^{N_ℓ} annihilates all of them, for every stratum and degree. The same power annihilates the corresponding locally constant cohomology sheaves. Choose d bounding the number of their possibly nonzero cohomological degrees. The finite Postnikov filtration then shows that ℓ^{dN_ℓ} annihilates each complex $\tau^{>c_b} G_b$. Thus the truncation maps become isomorphisms after rationalization [BH22, Remark 4.9].

At each step of the chosen finite filtration of $\mathcal{V} = \operatorname{Spec} A_V$ by closed subschemes, write j for the open stratum and i for its closed complement. The localization triangle is $j_! j^* G_\ell \to G_\ell \to i_* i^* G_\ell \to j_! j^* G_\ell[1]$. Both $j_!$ and i_* preserve scalar annihilators and upper perverse bounds. Induction along the finite filtration therefore gives (3.6) after rationalization. Conversely, successive $*$-restrictions preserve upper perverse bounds, and the shifted ordinary bound on each locally constant restriction gives (3.7). By [BH22, Theorem 4.11], affinoid comparison identifies the resulting localized structure with the rational perverse structure on the whole affinoid V.

On S_b, the rational cohomology sheaf $\mathcal{H}^q(F_\ell)$ is locally constant with fiber $M^q_{\ell,b}[1/\ell]$, so it is either zero everywhere or nonzero everywhere. Consequently,

$$\dim \overline{\operatorname{supp} \mathcal{H}^q(F_\ell)} = \max_{M^q_{\ell,b}[1/\ell] \neq 0} e_b,$$

with value $-\infty$ for the empty support. This identifies (3.7) with (3.8) and proves the lemma. □

Lemma 3.5. *Use the notation of Lemma 3.3. Let $\gamma : V \to S^{\mathrm{an}}$ be an analytic graph obtained by r successive cuts on a qcqs neighborhood over K. For corresponding objects E_ℓ in (3.5), form their restricted analytifications on this neighborhood as in Section 2.1, and denote them by the same symbols. The upper perverse bounds of $\gamma^* E_\ell[-r]$ are independent of ℓ. This includes the strict upper bounds after restriction to the corresponding closed boundaries.*

Proof. Work on an affinoid chart $V = \operatorname{Spa}(A_V)$ whose graph maps into an affine algebraic open of S. A classical point $x \in V$ has residue field finite over K. Its image $x_0 \in S$ is an algebraic closed point: evaluation of a K-algebra of finite type in $\kappa(x)$ has finite-dimensional image, which is a field. Ordinary pullback preserves geometric stalks, so the stalk cohomology of $\gamma^* E_\ell[-r]$ at x is the algebraic stalk cohomology at x_0, shifted by $-r$. Lemma 3.3 makes these dimensions independent of ℓ.

Compare two coefficient primes. Choose integral algebraic models for the finitely many corresponding objects, pull them back to $\operatorname{Spec} A_V$, and include the shift $[-r]$. Lemma 3.4 gives a common finite regular stratification with perfect values. Since A_V is Jacobson, every nonempty stratum contains closed points of $\operatorname{Spec} A_V$, corresponding to classical points of V. Rational cohomology ranks are constant on each stratum. Equality at classical points therefore shows that the same strata support nonzero cohomology in each degree for both primes. The closures of their supports agree and therefore have the same dimension, so (3.8) gives identical upper perverse

bounds. The same argument applies to the two boundary complexes in (3.5), using closed base change, and hence also to their strict upper bounds. □

Theorem 3.6. *The conclusions of Theorems 3.1 and 3.2 remain valid when $\ell = p$. In particular, every proper morphism with smooth source and a global relatively ample line bundle has the decomposition in (1.1) for every prime.*

Proof. We work locally on the base as in Theorems 3.1 and 3.2.

Step 1. Fix one geometric approximation. Choose an auxiliary prime $\ell_0 \neq p$, and apply Proposition 2.6 to the projective morphism and a prescribed closed boundary. Keep this algebraic model, its finite diagram resolving the inverse image of the boundary, and its analytic graph fixed for all coefficient primes. Write $g_0 : W_0 \to S$ and $i_0 : Z_0 \hookrightarrow S$ for the algebraic morphism and boundary, and use $K_{0,\ell}, P_{a,\ell}, I_{a,\ell}$ as in Lemma 3.3. Put $Q_{a,p} = {}^p\mathcal{H}^0(i_0^! P_{a,p})$. Let $M_0 \subset S^{\mathrm{an}}$ be the chosen qcqs neighborhood and $\gamma : V \hookrightarrow M_0$ the graph immersion. We use the same notation for the restricted analytifications. Let r be the codimension of γ, put $T^{(r)} = \gamma^*[-r]$, and write $i : Z \hookrightarrow V$ for the prescribed boundary.

Step 2. Transfer the upper perverse bounds. For $\ell_0 \neq p$, the proofs of Theorems 3.1 and 3.2 show that $T^{(r)}P_{a,\ell_0}$ and $T^{(r)}I_{a,\ell_0}$ are perverse. They also give the strict boundary bounds for $T^{(r)}I_{a,\ell_0}$ and its Verdier dual. Equation (2.10) identifies these duals, up to Tate twist, with the algebraic duals restricted to the graph and shifted by $[-r]$. Lemma 3.5 therefore transfers all six upper bounds in (3.5) to p. In particular, $T^{(r)}P_{a,p}$, $T^{(r)}I_{a,p}$, and the restrictions of their algebraic duals belong to ${}^pD^{\leq 0}$, while the two transported boundary complexes belong to ${}^pD^{\leq -1}$.

Step 3. Obtain the lower bounds by Gysin duality. Algebraic Lefschetz splitting and the algebraic decomposition by supports make $P_{a,p}$, $I_{a,p}$, and their Verdier duals shifted and Tate-twisted derived retracts of $K_{0,p}$. Pullback preserves these retractions, and proper base change identifies the ambient object after each cut with the normalized direct image from the new smooth source. Thus Lemma 2.9 applies at every stage. For $E = P_{a,p}$ or $I_{a,p}$, equation (2.10) gives

$$\mathbb{D}(T^{(r)}E) \simeq T^{(r)}(\mathbb{D}E)(-r).$$

The right-hand side belongs to ${}^pD^{\leq 0}$ by Step 2. Verdier duality with rational coefficients gives $T^{(r)}E \in {}^pD^{\geq 0}$, so both $T^{(r)}P_{a,p}$ and $T^{(r)}I_{a,p}$ are perverse.

Step 4. Preserve the strict boundary conditions. Let $\gamma_Z : Z \to Z_0$ be the map induced by γ, with Z_0 restricted to M_0. For each coefficient prime, closed base change gives the natural identities $\gamma^* i_{0*} \simeq i_* \gamma_Z^*$ and $\gamma_Z^* i_0^* = i^* \gamma^*$. These hold for integral complexes and commute with rationalization [BH22, Theorems 3.15 and 3.36]. Applying them to $I_{a,p}$ and $\mathbb{D}I_{a,p}$, the transferred bounds give

$$i_* i^* T^{(r)} I_{a,p} \simeq T^{(r)}(i_{0*} i_0^* I_{a,p}) \in {}^pD^{\leq -1},$$
$$i_* i^* \mathbb{D}(T^{(r)} I_{a,p}) \simeq T^{(r)}(i_{0*} i_0^* \mathbb{D} I_{a,p})(-r) \in {}^pD^{\leq -1},$$

where the second identification uses (2.10). Hence $T^{(r)}I_{a,p}$ is the intermediate extension of its open restriction. The algebraic decomposition $P_{a,p} \simeq I_{a,p} \oplus i_{0*}Q_{a,p}$ remains a direct-sum decomposition after restriction to the graph. Its complementary summand is perverse, as a derived summand of the perverse object $T^{(r)}P_{a,p}$, and is supported on the final boundary. Lemma 2.2 therefore makes the canonical boundary map for $T^{(r)}P_{a,p}$ an isomorphism.

Step 5. Recover hard Lefschetz and the global decomposition. Algebraic relative hard Lefschetz applies to $K_{0,p}$, since the algebraic model has characteristic zero. The restrictions of its perverse summands to the graph are now known to be perverse, so pullback of the algebraic splitting identifies them with the perverse cohomology of the rigid direct image, as in (3.2). The matrix argument of Theorem 3.1 identifies the transported Lefschetz maps with those induced by the original relatively ample line bundle, up to the invertible scalar arising from a tensor power. These canonical maps are isomorphisms locally over K, hence globally. Lemma 2.1 gives the global derived splitting. The boundary argument applies to every prescribed closed subset, and Corollary 2.3 decomposes each perverse cohomology sheaf by supports. □

3.4. **Singular sources and derived retracts.** We resolve the source and realize its intersection complex as a derived direct summand of the direct image of the constant sheaf. The next lemma shows that this summand inherits the decomposition.

Lemma 3.7. *Let M be a qcqs rigid space, and suppose $E \simeq \bigoplus_a P_a[-a]$ is a finite direct sum in $D^b_{\mathrm{zc}}(M, \mathbf{Q}_\ell)$, with each P_a a finite direct sum of intermediate extensions of finite-rank local systems on smooth Zariski-locally closed strata. Every derived direct summand of E has a decomposition of the same form.*

Proof. For $a < b$, the perverse t-structure gives

$$\mathrm{Hom}(P_a[-a], P_b[-b]) = \mathrm{Hom}(P_a, P_b[a-b]) = 0.$$

Thus $\mathrm{End}(E)$ is triangular, and its strictly triangular ideal J is nilpotent because only finitely many degrees occur. Let $e \in \mathrm{End}(E)$ satisfy $e^2 = e$ and define the chosen derived retract. Write $e_{aa} : P_a \to P_a$ for its diagonal components and put $d = \bigoplus_a e_{aa}[-a]$. Triangularity gives $e_{aa}^2 = e_{aa}$, so $d^2 = d$. Set $u = ed + (1-e)(1-d)$. Then $u \equiv 1 \pmod J$, hence u is invertible, and $eu = ud$. It follows that $e = udu^{-1}$, so its image is isomorphic to $\bigoplus_a \mathrm{im}(e_{aa})[-a]$. These diagonal images exist in the abelian perverse heart.

To describe each diagonal image, group the summands of P_a given by intermediate extension according to their irreducible supports. There are no nonzero morphisms between summands with distinct strict supports: the image would be both a quotient of one and a subobject of the other, supported on a proper closed subset of at least one support. Thus e_{aa} preserves each group. On a fixed support Z, choose a common smooth dense open U on which all the summands are shifted local systems. Full faithfulness of intermediate extension identifies e_{aa} on this group with an idempotent of the direct sum of these local systems on U. Its image is a finite-rank local system, and intermediate extension preserves the resulting direct-sum decomposition. Hence each $\mathrm{im}(e_{aa})$ is a finite direct sum of intermediate extensions, proving the lemma. □

Proof of Theorem 1.1. Let $f : X \to Y$ be projective, with Y qcqs. Then X is qcqs. Nilpotent thickenings do not change the étale objects under consideration, and the componentwise definition of IC_X allows us to treat the finitely many reduced irreducible components separately, shifting by the dimension of each component and applying its closed pushforward. We may therefore assume that X is reduced and pure-dimensional of dimension d.

Step 1. Resolve the source. Temkin's analytic desingularization theorem [Tem12, Section 5.2.1 and Theorem 5.2.2] gives a resolution $\pi : \widetilde{X} \to X$ by analytic blow-ups, with $\widetilde{X}$ smooth of pure dimension d and π an isomorphism over a smooth dense open $j : U \hookrightarrow X$. Quasi-compactness makes the sequence finite. Each blow-up is projective and carries a global relatively ample tautological line bundle. For proper maps $h : V \to W$ and $g : W \to S$ with relatively ample line bundles A and B, respectively, $A \otimes h^*B^{\otimes n}$ is $g \circ h$-ample for sufficiently large n when S is qcqs. Locally on S, relative GAGA and the relative Proj construction [Con06, Example 3.2.6 and Theorem 3.2.7] reduce this to the algebraic composition theorem [Stacks, Tag 0C4K]; a finite affinoid covering gives one exponent on S. Applying this successively gives global relatively ample line bundles for π and $f \circ \pi$. Theorem 3.6 therefore applies to both maps.

Step 2. Realize the intersection complex as a retract. Put $Z = X \setminus U$, $A = R\pi_*\mathbf{Q}_\ell[d]$, and $P = {}^p\mathcal{H}^0(A)$. Since π is an isomorphism over U, we have $P|_U \simeq \mathbf{Q}_\ell[d]$. The boundary splitting supplied by Theorem 3.6 gives $P \simeq j_{!*}(\mathbf{Q}_\ell[d]) \oplus P_Z$, where P_Z is supported on $X \setminus U$. By definition, the first summand is IC_X. The global derived splitting for π makes P a derived summand of A. Composing the perverse and derived retractions gives

$$\mathrm{IC}_X \longrightarrow R\pi_*\mathbf{Q}_\ell[d] \longrightarrow \mathrm{IC}_X$$

with composite $\mathrm{id}_{\mathrm{IC}_X}$.

Step 3. Apply the projective direct image. Applying Rf_* shows that $Rf_*\,\mathrm{IC}_X$ is a derived retract of $R(f\circ\pi)_*\mathbf{Q}_\ell[d]$. The source $\widetilde{X}$ is smooth, so Theorem 3.6 decomposes this latter object into finitely many shifts of intermediate extensions of local systems. Lemma 3.7 gives a decomposition of the same form for $Rf_*\,\mathrm{IC}_X$, proving (1.1). □

Corollary 3.8. *Let $f : X \to Y$ be as in Theorem 1.1, and put $C = \widehat{\overline{K}}$. For every prime ℓ, including p, the perverse Leray spectral sequence*

$$E_2^{a,b} = H^a\big(Y_C, ({}^p\mathcal{H}^b(Rf_*\,\mathrm{IC}_X))_C\big) \Longrightarrow H^{a+b}\big(X_C, (\mathrm{IC}_X)_C\big)$$

degenerates at E_2.

Proof. Base change the derived decomposition in (1.1) to C and apply $R\Gamma(Y_C, -)$. The perverse truncation filtration is split, so every differential from E_2 onwards vanishes. □

4. De Rham intersection cohomology

Theorem 4.3 and its consequences through Corollary 4.12 use comparison theorems, normalization, and finite covers, independently of projective decomposition. In Section 4.3, we use the retraction obtained from a resolution and the decomposition for smooth sources, both from Section 3, to prove [BH22, Conjecture 4.16(1)] for proper spaces. They also give de Rham intersection cohomology for local systems that extend across the boundary or admit a projective realization.

Throughout this section, $K/\mathbf{Q}_p$ is finite, $C = \widehat{\overline{K}}$, and $G_K = \mathrm{Gal}(\overline{K}/K)$. Let X/K be proper, reduced, and pure-dimensional of dimension d. Let $j : U \hookrightarrow X$ be the inclusion of a smooth dense Zariski-open subspace, and let $\mathbb{L}$ be a de Rham $\mathbf{Z}_p$-local system on U. Write

$$L = \mathbb{L} \otimes_{\mathbf{Z}_p} \mathbf{Q}_p, \qquad P(L) = \mathrm{IC}_X(L) = j_{!*}(L[d]), \qquad \mathrm{IH}^m(X_C, L) = H^m(X_C, P(L)_C).$$

This is the perverse normalization of [BH22, Construction 4.12]. In particular, when X is smooth and L is defined on all of X, $\mathrm{IH}^m(X_C, L) = H^{m+d}(X_C, L)$. The de Rham assertions concern the rational Galois representations associated with the given integral lattices. For a nonproper smooth open we take integral cohomology before inverting p. More explicitly, with $\mathbb{L}_n = \mathbb{L}/p^n$, our conventions are

$$R\Gamma(U_C, L) = \big(R\varprojlim_n R\Gamma(U_C, \mathbb{L}_n)\big)[1/p],$$
$$R\Gamma_c(U_C, L) = \big(R\varprojlim_n R\Gamma(X_C, j_{C,!}\mathbb{L}_n)\big)[1/p].$$

These are the conventions used by the logarithmic comparison theorems; see also [LLZ23, Lemma 2.4.3 and Remark 2.4.4]. Direct images along open immersions are likewise formed from the integral system on the fixed proper ambient space and then rationalized.

4.1. **The boundary range.** The dimension of the boundary bounds the degrees in which intersection cohomology can differ from ordinary or compactly supported cohomology of the smooth open subset.

Lemma 4.1. *For every integer q, the representations $H^q(U_C, L)$ and $H^q_c(U_C, L)$ are finite-dimensional de Rham representations of G_K.*

Proof. Apply Temkin's desingularization and weak principalization theorems to the pair $(X, X \setminus U)$ [Tem17, Theorems 1.1.11 and 1.1.13(i)]. Since U is smooth, the resulting proper modification $\pi : \widetilde{X} \to X$ is an isomorphism over U, with $\widetilde{X}$ smooth and $D = (\widetilde{X} \setminus U)_{\mathrm{red}}$ a strict normal crossings divisor. The space $\widetilde{X}$ is proper because X is proper. Equip $\widetilde{X}$ with the log structure defined by D. By [DLLZa, Corollary 6.3.4], the local system $\mathbb{L}$ extends to a Kummer étale $\mathbf{Z}_p$-local system on this log space. The logarithmic de Rham comparison theorem [DLLZb, Corollary 3.2.10] then gives a finite-dimensional, filtered, G_K-equivariant comparison over Fontaine's de Rham period field B_{dR} for $H^q(U_C, L)$, so this representation is de Rham.

For compactly supported cohomology, apply [LLZ23, Theorem 3.1.10] with compact supports imposed along every component of D. This gives the corresponding finite-dimensional de Rham comparison. Writing $\widetilde{j} : U \hookrightarrow \widetilde{X}$ and $\mathbb{L}_n = \mathbb{L}/p^n$, proper base change gives $R\pi_*\widetilde{j}_!\mathbb{L}_n \simeq j_!\mathbb{L}_n$: over U the map π is an isomorphism, and over $X \setminus U$ both sides have zero stalks. These identifications are compatible with reduction in n. Taking derived inverse limits and then inverting p gives the cohomology groups with compact support defined above. □

Lemma 4.2. *Let S/K be proper of dimension at most b, and let $F \in {}^pD^{\le -1}(S, \mathbf{Q}_p)$. Then $R\Gamma(S_C, F_C) \in D^{\le b-1}(\mathbf{Q}_p)$.*

Proof. Choose an integral model A_0 of F, and put $A = {}^p\tau_{\le -1}A_0$. Rationalization commutes with perverse truncation, so $A[1/p] \simeq F$. For every $n \ge 1$, the reduction $A_n = A \otimes^{\mathbf{L}}_{\mathbf{Z}_p} \mathbf{Z}/p^n$ lies in ${}^pD^{\le -1}(S, \mathbf{Z}/p^n)$ by [BH22, Proposition 4.6(2) and Theorem 4.11]. For each nonzero $\mathcal{H}^q(A_n)_C$, let Z_q be the Zariski closure of its support with the reduced structure, let $a_q : Z_q \hookrightarrow S_C$ be the inclusion, and put $s_q = \dim Z_q$. The perverse support condition gives $q + s_q \le -1$, and $s_q \le b$. Since $\mathcal{H}^q(A_n)_C \simeq a_{q*}a_q^*\mathcal{H}^q(A_n)_C$ and closed pushforward is exact, its cohomology can be computed on Z_q. The bound on étale cohomological dimension for torsion sheaves on this qcqs space over C, including for p-torsion, gives $H^r(S_C, \mathcal{H}^q(A_n)_C) = 0$ for $r > 2s_q$ [Ber93, Theorem 4.2.6]; see also [Han17, Introduction, p. 2] for its formulation in rigid geometry. Thus every nonzero term in the ordinary hypercohomology spectral sequence has total degree bounded by

$$q + r \le q + 2s_q \le s_q - 1 \le b - 1.$$

It follows that $R\Gamma(S_C,(A_n)_C)\in D^{\le b-1}$ for every n. Proper finiteness [BH22, Theorem 3.10] makes the cohomology groups for each n finite, so the inverse systems in each degree satisfy the Mittag–Leffler condition. In particular, $R^1\varprojlim_n H^{b-1}(S_C,(A_n)_C)=0$, so the derived inverse limit has no cohomology above degree $b-1$. Inverting p proves the assertion. □

Theorem 4.3. *Suppose $Z=X\setminus U$ is nonempty, and put $b=\dim Z$. Then $\mathrm{IH}^m(X_C,L)$ is de Rham whenever*

$$m\le -b \qquad or \qquad m\ge b. \tag{4.1}$$

More precisely, the canonical restriction map

$$\mathrm{IH}^m(X_C,L)\longrightarrow H^{m+d}(U_C,L) \tag{4.2}$$

is an isomorphism for $m<-b$ and injective for $m=-b$. The canonical map from compactly supported cohomology

$$H_c^{m+d}(U_C,L)\longrightarrow \mathrm{IH}^m(X_C,L) \tag{4.3}$$

is an isomorphism for $m>b$ and surjective for $m=b$. If Z is empty, all intersection cohomology groups are de Rham.

Proof. Let $i:Z\hookrightarrow X$, and write $P=P(L)$. The strict cosupport condition for intermediate extension gives $i^!P\in {}^pD^{\ge 1}(Z,\mathbf{Q}_p)$. On a space of dimension at most b, the support definition of the perverse t-structure gives $D^{\le -b}\subset {}^pD^{\le 0}$: every possibly nonzero ordinary cohomology sheaf on the left has degree $q\le -b$ and support of dimension at most $b\le -q$. Taking right orthogonals yields

$${}^pD^{\ge 1}\subset D^{\ge 1-b}. \tag{4.4}$$

Here we use the rational perverse structure of [BH22, Definition 4.1, Proposition 4.6, and Theorem 4.11]. After base change to C, left t-exactness of derived global sections for the ordinary t-structure gives $R\Gamma(Z_C,i_C^!P_C)\in D^{\ge 1-b}(\mathbf{Q}_p)$. Consider the localization triangle

$$i_*i^!P\longrightarrow P\longrightarrow Rj_*L[d]\longrightarrow i_*i^!P[1]. \tag{4.5}$$

Its long exact sequence gives the isomorphism and injection in (4.2). We form $Rj_*L[d]$ from the given integral lattice on the proper ambient space X, using [BH22, Theorem 3.36(4)], and then invert p; its geometric cohomology is therefore the cohomology of U_C defined above. All maps are defined over K, so the resulting maps of representations are G_K-equivariant. Lemma 4.1 and stability of de Rham representations under subrepresentations prove the assertion for $m\le -b$.

For the upper range, the strict support condition gives $i^*P\in {}^pD^{\le -1}(Z,\mathbf{Q}_p)$, and Lemma 4.2 gives $R\Gamma(Z_C,i_C^*P_C)\in D^{\le b-1}(\mathbf{Q}_p)$. The localization triangle

$$j_!L[d]\longrightarrow P\longrightarrow i_*i^*P\longrightarrow j_!L[d+1]$$

then gives the isomorphism and surjection in (4.3). Its source is de Rham by Lemma 4.1, and de Rham representations are closed under quotients. This proves the assertion for $m\ge b$. If Z is empty, $X=U$ is smooth and proper, so smooth proper de Rham comparison [Sch13, Theorem 1.6] applies. □

Remark 4.4. Smooth duality [LRZ24, Corollary 7.1.2], compatibility of intermediate extension with Verdier duality, and Lemma 2.10 give

$$\mathbb{D}_X P(L)\simeq P(L^\vee)(d), \tag{4.6}$$

$$\mathrm{IH}^m(X_C,L)\simeq \mathrm{IH}^{-m}(X_C,L^\vee)^\vee(-d). \tag{4.7}$$

Thus duality exchanges the lower and upper ranges in Theorem 4.3.

Corollary 4.5. *If $X\setminus U$ is finite, every $\mathrm{IH}^m(X_C,L)$ is de Rham. In particular, this holds for a proper reduced curve with de Rham local coefficients on a smooth dense open subset, and for constant coefficients on a proper pure-dimensional rigid space with finite singular locus.*

Proof. For nonempty finite boundary, $b=0$, and the two ranges in (4.1) cover every integer. The case of an empty boundary is included in Theorem 4.3. The complement of a dense Zariski open in a proper curve is finite; for constant coefficients, take $U=X^{\mathrm{sm}}$. □

Proposition 4.6. *Let $i:Z=X\setminus U\hookrightarrow X$, and let $T\subset Z$ be closed. Suppose that the canonical morphism $j_!L\to Rj_*L$, formed on X using the given lattice, is an isomorphism on $X\setminus T$. If T is nonempty of dimension c, then $\mathrm{IH}^m(X_C,L)$ is de Rham for $m\le -c$ or $m\ge c$. If T is finite or empty, it is de Rham in every degree.*

Proof. On $X \setminus T$, the complexes $j_!L[d]$ and $Rj_*L[d]$ are isomorphic. The first has the upper perverse bound and the second the lower perverse bound, so their common restriction is perverse. Its $*$-restriction to the boundary vanishes because it is an extension by zero, and its !-restriction to the boundary vanishes because it is a direct image under the open immersion. It is therefore the intermediate extension there, and both $i^*P(L)$ and $i^!P(L)$ are supported on T. These assertions may be checked on qcqs affinoid neighborhoods in $X \setminus T$, with the complexes formed on X.

Closed pushforward is perverse t-exact and reflects perverse bounds for objects supported on its image. We may therefore apply the argument of Theorem 4.3 to the boundary complexes on T, using (4.4) and Lemma 4.2 with c in place of b. If T is finite, the resulting ranges cover every degree. If T is empty, $P(L) = Rj_*L[d]$, and Lemma 4.1 gives the assertion in every degree. □

Under the hypotheses of Proposition 4.6, assume that T is finite. The two localization maps then identify the middle intersection cohomology with

$$\mathrm{IH}^0(X_C, L) \simeq \mathrm{im}\big(H^d_c(U_C, L) \longrightarrow H^d(U_C, L)\big). \tag{4.8}$$

Indeed, the boundary complexes are supported at finitely many points, where their strict perverse bounds become ordinary cohomological bounds. Geometric global sections on these supports are exact, so the localization maps give a surjection from $H^d_c(U_C, L)$ onto $\mathrm{IH}^0(X_C, L)$ and an injection from $\mathrm{IH}^0(X_C, L)$ into $H^d(U_C, L)$. Their composite is the usual map from compactly supported to ordinary cohomology.

4.2. **Normalization and finite covers.** Normalization and finite covers allow us to apply Theorem 4.3 with a smaller boundary. We use the compatibility of finite pushforward with the perverse t-structure and intermediate extension.

Proposition 4.7. *Let $\nu : X^\nu \to X$ be the disjoint union of the normalizations of the reduced irreducible components, and put $W = (X^\nu)^{\mathrm{sm}}$. Suppose that the pullback of $\mathbb{L}$ from $U \simeq \nu^{-1}U$ extends to a $\mathbf{Z}_p$-local system $\mathbb{M}$ on W. If $X^\nu \setminus W$ is nonempty of dimension b_ν, then $\mathrm{IH}^m(X_C, L)$ is de Rham for $m \le -b_\nu$ or $m \ge b_\nu$. If X^ν is smooth, it is de Rham in every degree. Under the stated extension hypothesis, the same conclusion holds when $\dim X \le 2$.*

Proof. Normalization is finite because affinoid algebras are excellent. The local system $M = \mathbb{M}[1/p]$ agrees with the pullback of L on the dense open $\nu^{-1}U$. Every connected smooth affinoid open in W meets this subset, so Liu–Zhu rigidity [LZ16, Theorem 3.9(iv)] implies that M is de Rham on W. Let $a : W \hookrightarrow X^\nu$. Intermediate extension is unchanged when the smooth dense open is shrunk, and finite pushforward is perverse t-exact and preserves the strict boundary conditions. Consequently,

$$P(L) \simeq \nu_* a_{!*}(M[d]).$$

The assertion on W is local on smooth affinoids, and both intersection complexes are formed on proper qcqs ambient spaces.

Finite pushforward introduces no degree shift in geometric cohomology. Applying Theorem 4.3 to $(X^\nu, W, \mathbb{M})$ gives the stated ranges, including every degree when X^ν is smooth. Since K has characteristic zero, a normal rigid space over K is smooth in codimension one. Thus normal curves and reduced zero-dimensional spaces are smooth, while the singular locus of a normal surface has dimension at most zero. This proves the conclusion in dimensions at most two. □

Corollary 4.8. *If X has pure dimension $d \ge 2$, then $\mathrm{IH}^m(X_C, \mathbf{Q}_p)$ is de Rham for $m \le 2-d$ or $m \ge d-2$. In particular, every intersection cohomology group of a proper rigid space of dimension at most two is de Rham.*

Proof. Apply Proposition 4.7 to the constant local system on $(X^\nu)^{\mathrm{sm}}$. Its complement has dimension at most $d-2$, giving the stated ranges, which cover every integer for $d = 2$. The same proposition handles the lower dimensions. Finite sums and invariance under nilpotent thickenings give the last assertion. □

Proposition 4.9. *Suppose that $\mathbb{L}$ has finite geometric monodromy on each geometrically connected component of U. If $d \ge 2$, then $\mathrm{IH}^m(X_C, L)$ is de Rham for $m \le 2-d$ or $m \ge d-2$. If $d \le 2$, it is de Rham in every degree.*

Proof. Finite normalization and the characterization of intermediate extension by strict conditions on the boundary reduce us to normal X. We may also work componentwise after a finite extension of K, since intermediate extension commutes with this base change and the de Rham property is detected after a finite field extension. Put $r = \mathrm{rank}\,\mathbb{L}$. Choose a geometric base point, let $\rho : \pi_1(U) \to \mathrm{GL}_r(\mathbf{Z}_p)$ be the monodromy representation,

and put $H = \rho(\pi_1(U_C))$, which is finite by hypothesis. Choose n such that reduction modulo p^n is injective on H. The finite étale frame cover $g : U' \to U$ associated with the kernel of $\rho \bmod p^n$ kills the geometric monodromy: a geometric element acting trivially modulo p^n already acts trivially on $\mathbb{L}$. Hansen's extension theorem [Han20, Theorem 1.6] extends g to a finite cover $f : Y \to X$ with Y normal. The space Y is proper and pure of dimension d.

After a further finite extension of K, the components Y_α and their dense smooth opens U'_α are geometrically connected. The pullback $L'_\alpha = g^*L|_{U'_\alpha}$ is geometrically constant, so evaluation identifies it with the arithmetic constant local system associated with $V_\alpha = H^0(U'_{\alpha,C}, L'_\alpha)$. This identification is G_K-equivariant. Since L'_α is de Rham, Lemma 4.1, applied to $U'_\alpha \subset Y_\alpha$, shows that V_α is de Rham.

The unit and the trace divided by the degree exhibit L as a rational direct summand of g_*g^*L. Intermediate extension preserves this split pair, finite pushforward commutes with intermediate extension, and tensoring by V_α also commutes with intermediate extension. Hence $P(L)$ is a direct summand of $f_* \bigoplus_\alpha (\mathrm{IC}_{Y_\alpha,\mathbf{Q}_p} \otimes_{\mathbf{Q}_p} V_\alpha)$. Taking geometric cohomology realizes $\mathrm{IH}^m(X_C, L)$ as a direct summand of

$$\bigoplus_\alpha \mathrm{IH}^m(Y_{\alpha,C}, \mathbf{Q}_p) \otimes_{\mathbf{Q}_p} V_\alpha.$$

Corollary 4.8 gives the required ranges for each factor with constant coefficients. Stability of de Rham representations under tensor products, finite direct sums, and direct summands, followed by descent along the finite field extension, proves the result. □

Definition 4.10. Identify U with its inverse image in the componentwise normalization X^ν. Choose a closed analytic subset $T \subset X^\nu \setminus U$ of codimension at least two such that $X^\circ = X^\nu \setminus T$ is smooth and $D = X^\circ \setminus U$ is a disjoint union of smooth divisors. Such a choice is obtained by removing the singular locus of X^ν, the singular loci and pairwise intersections of its divisorial boundary components, and the boundary components of higher codimension. Equip X° with the log structure defined by D. For a geometric point ξ of D and a log-geometric point $\widetilde{\xi}$ above it, let $X^\circ(\xi)$ be the strict localization at ξ, with the log structure pulled back from X°. Writing $\pi_1^{\text{két}}$ for the Kummer étale fundamental group, the geometric Kummer inertia group is

$$I_\xi = \pi_1^{\text{két}}(X^\circ(\xi), \widetilde{\xi}) \simeq \widehat{\mathbf{Z}}(1)$$

[DLLZa, Proposition 4.4.9, Definition 6.3.7, and Example 6.3.8]. We say that $\mathbb{L}$ has finite divisorial inertia if, for one such choice of T, every I_ξ has finite image on the corresponding stalk of its Kummer étale extension.

For a smooth logarithmic pair (Y, D), write $Y_{\text{két}}$ and $Y_{\text{ét}}$ for its Kummer étale and ordinary étale sites. We use the following local descent criterion: a finite locally constant sheaf on $Y_{\text{két}}$ comes from $Y_{\text{ét}}$ if and only if every geometric Kummer inertia group acts trivially. Indeed, represent it by a finite Kummer étale cover. Over a strict localization, trivial inertia identifies this cover with a disjoint union of copies of the base [DLLZa, Proposition 4.4.9 and Corollary 4.4.22], so its underlying finite morphism is étale. Full faithfulness [DLLZa, Remark 4.4.23] gives uniqueness and preserves morphisms, module structures, and transition maps.

Proposition 4.11. *If $\mathbb{L}$ has finite divisorial inertia, then $\mathrm{IH}^m(X_C, L)$ is de Rham for $m \le 2-d$ or $m \ge d-2$ when $d \ge 2$, and in every degree when $d \le 2$.*

Proof. Finite normalization reduces us to normal X. Take the finite étale frame cover $g : U' \to U$ that trivializes $\mathbb{L}/p^2$, and extend it to a finite normal cover $f : Y \to X$ by [Han20, Theorem 1.6]. On each connected component of U', the monodromy of $\mathbb{L}' = g^*\mathbb{L}$ takes values in $1 + p^2 M_r(\mathbf{Z}_p)$, where r is the rank of $\mathbb{L}'$ on that component. This group is torsion-free for every prime p, including $p = 2$: the matrix logarithm converges and is injective there, and an element of finite order has zero logarithm.

Choose X°, D, T as in Definition 4.10. Rigid Abhyankar's lemma [DLLZa, Proposition 4.2.1] identifies $Y^\circ = f^{-1}(X^\circ)$ as a finite Kummer étale cover of (X°, D). Since D is a disjoint union of smooth divisors, its log rank is at most one. Up to étale maps, the local Kummer charts have the form $t = s^e$, so Y° is smooth and its boundary is again a disjoint union of smooth divisors. The geometric inertia groups on Y° map into those on X°. Their images under $\mathbb{L}'$ are finite by hypothesis and lie in the torsion-free congruence subgroup, so they are trivial. The local descent criterion therefore extends each $\mathbb{L}'/p^n$ uniquely as an ordinary étale local system on Y°, compatibly with its module structure and the transition maps.

Put $W = Y^{\mathrm{sm}}$. The closed subset $W \setminus Y^\circ$ is contained in $f^{-1}T$ and has codimension at least two. Rigid analytic purity [Han20, Corollary 2.15] extends every $\mathbb{L}'/p^n$ uniquely to W. Full faithfulness extends the module structures and transition maps, so these extensions for varying n form a $\mathbf{Z}_p$-local system $\mathbb{M}$ on W. Liu–Zhu rigidity [LZ16, Theorem 3.9(iv)] makes $\mathbb{M}[1/p]$ de Rham on W, since it is de Rham on the dense open U'. Apply

Proposition 4.7 to each normal proper component of Y. For $d \geq 2$, its singular locus has dimension at most $d-2$; in dimensions at most two, the proposition gives every degree. Finally, the unit and normalized trace over U, followed by intermediate extension and finite pushforward, make $P(L)$ a direct summand of $f_* \mathrm{IC}_Y(g^*L)$. Taking geometric cohomology gives the asserted ranges. □

Corollary 4.12. *Every rank-one de Rham $\mathbf{Z}_p$-local system on U satisfies the conclusion of Proposition 4.11. In particular, all its intersection cohomology groups are de Rham when $\dim X \leq 2$, including when the local system is ramified along a divisorial boundary.*

Proof. We show that $\mathbb{L}$ has finite divisorial inertia. Choose X°, D, T as in Definition 4.10, and let x be a classical point of D with residue field K'/K finite. After passing to an étale neighborhood over K', choose a parameter defining D. The resulting split log point has Kummer fundamental group $I_x \rtimes G_{K'}$, where $I_x \simeq \widehat{\mathbf{Z}}(1)$ and $G_{K'}$ acts cyclotomically; see [DLLZa, Examples 4.4.24–4.4.25]. The rank-one local system gives a character $\rho : I_x \rtimes G_{K'} \to \mathbf{Z}_p^\times$. Let $\mu(\mathbf{Z}_p)$ be the group of roots of unity in $\mathbf{Z}_p^\times$, and put $w = |\mu(\mathbf{Z}_p)|$. The character ρ^w takes values in $1 + p\mathbf{Z}_p$, and in $1 + 4\mathbf{Z}_2$ when $p = 2$, where the p-adic logarithm is injective. Let $\chi_p : G_{K'} \to \mathbf{Z}_p^\times$ be the cyclotomic character. Since the target of ρ is abelian, the relation in the semidirect product gives $(\chi_p(\sigma) - 1)\log(\rho(t)^w) = 0$ for every $\sigma \in G_{K'}$ and every t in the pro-p subgroup of I_x. The cyclotomic character has infinite image, so $\rho(t)^w = 1$. The prime-to-p inertia has finite image in $\mathbf{Z}_p^\times$, hence also has image in $\mu(\mathbf{Z}_p)$. Thus $\rho(I_x) \subseteq \mu(\mathbf{Z}_p)$, and the inertia action on $\mathbb{L}^{\otimes w}$ is trivial at every classical point of the boundary.

For each n, consider the finite étale frame cover of $\mathbb{L}^{\otimes w}/p^n$ on U. Rigid Abhyankar's lemma [DLLZa, Proposition 4.2.1] extends it to a finite Kummer étale cover of (X°, D). The inertia is trivial at every classical point of the boundary, so the local descent criterion makes the extension ordinary étale there. A nonempty non-étale locus is a closed analytic subset and contains a classical point. Since the cover is already étale on U, this locus is empty. The extension is therefore finite étale everywhere on X°, and $\mathbb{L}^{\otimes w}/p^n$ has trivial geometric Kummer inertia at every log-geometric point of the boundary. This holds for all n, so $\mathbb{L}^{\otimes w}$ has trivial geometric Kummer inertia. The image of inertia acting on $\mathbb{L}$ consequently has exponent dividing w and is contained in the finite group $\mu(\mathbf{Z}_p)$. Proposition 4.11 now applies. □

4.3. **Consequences of projective decomposition.** With coefficients in $\mathbf{Q}_p$, the decomposition theorem of Section 3 realizes the intersection complex as a derived direct summand of the direct image from a smooth proper resolution. The retraction is defined over K, so geometric cohomology gives G_K-equivariant direct summands to which smooth proper de Rham comparison applies.

Corollary 4.13. *For every proper rigid space X/K and every integer m, the group $\mathrm{IH}^m(X_C, \mathbf{Q}_p)$ is a de Rham G_K-representation. Thus Theorem 1.1 implies the assertion for proper spaces in* [BH22, Conjecture 4.16(1)].

Proof. First suppose that X is reduced and pure-dimensional of dimension d. Choose a projective resolution $\pi : \widetilde{X} \to X$ over K as in Section 3.4. Its source is smooth and is proper over K because π and $X \to \mathrm{Spa}(K)$ are proper. The morphism π is an isomorphism over a smooth dense open subset. The decomposition with coefficients in $\mathbf{Q}_p$ and the splitting along the boundary used there give a derived retraction

$$\mathrm{IC}_{X,\mathbf{Q}_p} \xrightarrow{u} R\pi_*\mathbf{Q}_p[d] \xrightarrow{v} \mathrm{IC}_{X,\mathbf{Q}_p}, \qquad vu = \mathrm{id},$$

over K. After base change to C, proper base change identifies $R\Gamma(X_C, R\pi_{C*}\mathbf{Q}_p[d])$ with $R\Gamma(\widetilde{X}_C, \mathbf{Q}_p)[d]$. Taking cohomology in degree m gives a G_K-equivariant retraction

$$\mathrm{IH}^m(X_C, \mathbf{Q}_p) \longrightarrow H^{m+d}(\widetilde{X}_C, \mathbf{Q}_p) \longrightarrow \mathrm{IH}^m(X_C, \mathbf{Q}_p). \tag{4.9}$$

Scholze's smooth proper comparison theorem [Sch13, Corollary 1.8] makes the middle representation de Rham. Its direct summands are de Rham, proving the assertion in the pure-dimensional case.

For general X, nilpotent thickenings do not change the relevant étale objects. Apply the preceding argument to each reduced irreducible component, with its own dimension shift, and use the componentwise definition of the intersection complex. This also proves Corollary 1.2. □

If $\mathbb{L}$ has finite geometric monodromy on each geometrically connected component of U, then $\mathrm{IH}^m(X_C, L)$ is de Rham for every m, in every dimension. The proof of Proposition 4.9 realizes this intersection cohomology, after a finite extension K'/K, as a $G_{K'}$-equivariant direct summand of finite sums of $\mathrm{IH}^m(Y_{\alpha,C}, \mathbf{Q}_p) \otimes_{\mathbf{Q}_p} V_\alpha$, where each Y_α is proper and V_α is de Rham. Corollary 4.13 makes the first factor de Rham in every degree. Tensor products and direct summands preserve this property, which then descends to the original field.

Proposition 4.14. *Suppose that the de Rham $\mathbf{Z}_p$-local system $\mathbb{L}$ on U extends to a lisse $\mathbf{Z}_p$-local system $\mathbb{M}$ on X. Then $\mathrm{IH}^m(X_C, L)$ is de Rham for every integer m, where $L = \mathbb{L}[1/p]$.*

Proof. Put $M = \mathbb{M}[1/p]$, and choose the resolution $\pi : \widetilde{X} \to X$ used in the proof of Corollary 4.13. Tensoring with the finite-rank local system M preserves ordinary and perverse bounds, as well as the strict support and cosupport conditions for intermediate extension. Consequently, $M \otimes_{\mathbf{Q}_p} \mathrm{IC}_{X,\mathbf{Q}_p} \simeq P(L)$. Tensor the derived retraction in that corollary with M. The projection formula identifies $R\pi_*\mathbf{Q}_p[d] \otimes_{\mathbf{Q}_p} M$ with $R\pi_*(\pi^*M)[d]$, so $P(L)$ is a derived direct summand of $R\pi_*(\pi^*M)[d]$ over K.

Let $V \subseteq X$ be a dense open over which π is an isomorphism. The local system π^*M is de Rham on $\pi^{-1}(U \cap V)$. Every connected component of $\widetilde{X}$ meets this open, so Liu–Zhu rigidity [LZ16, Theorem 3.9(iv)] makes π^*M de Rham throughout $\widetilde{X}$. Since $\widetilde{X}$ is smooth and proper, comparison with de Rham local coefficients [Sch13, Theorem 1.6] shows that $H^{m+d}(\widetilde{X}_C, \pi^*M)$ is de Rham. Applying geometric cohomology to the derived retraction exhibits $\mathrm{IH}^m(X_C, L)$ as a G_K-equivariant direct summand of this representation. □

Proposition 4.15. *Let $g : W \to X$ be proper with a global relatively ample line bundle, and suppose that W is smooth of pure dimension N. Assume that $h : W_U = g^{-1}(U) \to U$ is smooth of pure relative dimension $e = N - d$. If L is a direct summand over K of $R^q h_*\mathbf{Q}_p(t)$ for some integers q, t, then $\mathrm{IH}^m(X_C, L)$ is de Rham for every integer m. More precisely, it is a G_K-equivariant direct summand of*

$$H^{m+q+d}(W_C, \mathbf{Q}_p)(t). \tag{4.10}$$

Proof. Put $A = Rg_*\mathbf{Q}_p[N]$, $a = q - e$, and $P_a = {}^p\mathcal{H}^a(A)$. Smooth proper base change makes the higher direct images of h lisse on U. Since U is smooth of dimension d, a local system shifted by $[d]$ is perverse. Perverse exactness of open restriction therefore gives $j^*P_a \simeq (R^q h_*\mathbf{Q}_p)[d]$. The given inclusion and retraction exhibit $L[d]$ as a direct summand of $j^*P_a(t)$, and intermediate extension preserves this splitting. The boundary splitting of Theorem 3.6 makes $j_{!*}j^*P_a(t)$ a perverse direct summand of $P_a(t)$, so $P(L)$ is a perverse direct summand of $P_a(t)$.

The global derived splitting in the same theorem makes $P_a[-a]$ a derived direct summand of A. Combining these retractions gives

$$P(L)[-a] \longrightarrow A(t) \longrightarrow P(L)[-a]$$

with composite the identity, all over K. Apply geometric global sections and take cohomology in degree $m + a$. The outer terms give $H^{m+a}(X_C, P(L)[-a]) = \mathrm{IH}^m(X_C, L)$, while the middle term gives $H^{m+a+N}(W_C, \mathbf{Q}_p)(t)$. Since $a = q - e$ and $e = N - d$, we have $m + a + N = m + q + d$, proving that $\mathrm{IH}^m(X_C, L)$ is a direct summand as stated in (4.10). The space W is proper over K, because both g and $X \to \mathrm{Spa}(K)$ are proper. Smooth proper comparison [Sch13, Corollary 1.8] makes the middle representation de Rham, and hence so is its G_K-equivariant direct summand. □

4.4. **Further questions.** The projective decomposition theorem gives the de Rham property in every degree for constant coefficients. The results of Section 4.3 also give this property for de Rham local systems that extend across the boundary, have finite geometric monodromy, or occur as direct summands of the cohomology of a smooth proper family admitting the stated projective realization. For arbitrary ramified de Rham coefficients with nonempty boundary, Theorem 4.3 leaves the middle range $-b < m < b$, where $b = \dim(X \setminus U)$.

For coefficients with $\ell \neq p$, Hansen–Zavyalov [HZ25, Theorems 1.2.1(1) and 3.2.8] establish the assertion about nearby cycles in [BH22, Conjecture 4.15(1)] in its corrected formulation, using the nearby-cycle functor with values in Deligne's topos. Their Remark 1.2.2 compares this functor with derived pushforward along specialization to the special fiber over a non-algebraically closed field. Lu–Zheng [LZ26, Corollary 1.3 and Remark 1.4] establish the nonnegativity of Frobenius weights predicted by [BH22, Conjecture 4.15(2)].

The conjecture that singular support is an analytic Lagrangian subset is formulated in [Zho25b, Conjecture 9.6]. Section 5.3 studies isotropic upper bounds for singular support with finite coefficients prime to p, together with the case of sheaves supported in dimension at most one.

5. Examples and further remarks

5.1. **Smooth families and restriction.**

Corollary 5.1. *Let $f : X \to Y$ be smooth projective of pure relative dimension d, where Y is qcqs, reduced, and pure-dimensional of dimension m. For every coefficient prime ℓ, the sheaves $L_q = R^q f_*\mathbf{Q}_\ell$ are lisse, and there*

are derived decompositions

$$Rf_*\mathbf{Q}_\ell \simeq \bigoplus_{q=0}^{2d} L_q[-q],$$
$$Rf_*\,\mathrm{IC}_X \simeq \bigoplus_{q=0}^{2d} \mathrm{IC}_Y(L_q|_{Y^{\mathrm{sm}}})[d-q].$$

Proof. Smooth proper base change makes each L_q lisse, including for $\ell = p$ by [LRZ24, Corollary 6.4.8] and passage from finite to rational coefficients. Let H be a relatively ample line bundle. Hard Lefschetz on the smooth projective geometric fibers, together with rigid GAGA and proper base change, gives isomorphisms $c_1(H)^a : L_{d-a} \xrightarrow{\sim} L_{d+a}(a)$ for $0 \le a \le d$. These are morphisms of local systems induced by the global Chern class, and their invertibility is checked on geometric fibers. Applying Lemma 2.1 to $Rf_*\mathbf{Q}_\ell[d]$ gives the first decomposition.

For $A \in D^b_{\mathrm{zc}}(Y, \mathbf{Q}_\ell)$, shifted smooth pullback is perverse t-exact and satisfies $\mathbb{D}_X(f^*A[d]) \simeq f^*(\mathbb{D}_Y A)[d](d)$, by smooth duality [LRZ24, Corollary 7.1.2], regular base change for internal Hom, and [BH22, Propositions 3.7 and 3.16]. Since Tate twists preserve perversity, shifted smooth pullback preserves the strict support and cosupport conditions for intermediate extension. Hence $\mathrm{IC}_X \simeq f^*\,\mathrm{IC}_Y[d]$. The projection formula and the first decomposition give

$$Rf_*\,\mathrm{IC}_X \simeq \mathrm{IC}_Y \otimes^{\mathbf{L}}_{\mathbf{Q}_\ell} Rf_*\mathbf{Q}_\ell[d] \simeq \bigoplus_q (\mathrm{IC}_Y \otimes_{\mathbf{Q}_\ell} L_q)[d-q].$$

Since L_q is lisse on all of Y, tensoring with it commutes with intermediate extension. Thus $\mathrm{IC}_Y \otimes_{\mathbf{Q}_\ell} L_q \simeq \mathrm{IC}_Y(L_q|_{Y^{\mathrm{sm}}})$, proving the second formula. □

Finite pushforward is perverse t-exact and commutes with intermediate extension in the corresponding Cartesian diagrams. Let $f : X \to Y$ be projective, with Y qcqs and X smooth of pure dimension n, and put $Y_r = \{y \in Y : \dim f^{-1}(y) \ge r\}$. The morphism f is semismall if $\dim Y_r \le n - 2r$ for every $r \ge 0$, and small if these inequalities are strict for $r > 0$. In the semismall case, proper base change and the bound on the cohomological dimension of the fibers give the perverse support inequalities for $Rf_*\mathbf{Q}_\ell[n]$, and proper duality gives the cosupport inequalities. Thus $Rf_*\mathbf{Q}_\ell[n] = Rf_*\,\mathrm{IC}_X$ is perverse, and its support decomposition follows from Theorem 3.6. For a small resolution f, the strict support and cosupport inequalities give $Rf_*\mathbf{Q}_\ell[n] \simeq \mathrm{IC}_Y$.

Example 5.2. Consider the algebraic threefold ordinary double point $Z = \{xy - zw = 0\} \subset \mathbb{A}^4_K$ and its small projective resolution

$$\pi : T = \mathrm{Tot}\big(\mathcal{O}_{\mathbf{P}^1}(-1)^{\oplus 2}\big) \longrightarrow Z.$$

In homogeneous coordinates $[s : t]$ on $\mathbf{P}^1$ and fiber coordinates u, v, the map is $x = su$, $z = sv$, $w = tu$, and $y = tv$. Smallness gives $\mathrm{IC}_Z \simeq R\pi_*\mathbf{Q}_\ell[3]$. Take $h = x + y$. Its pullback is $h \circ \pi = su + tv$, whose differential in the fiber directions has components s, t. These never vanish simultaneously, so $h \circ \pi$ is smooth. For $\ell \ne p$, smooth local acyclicity and proper pushforward give $\phi_h(R\pi_*\mathbf{Z}/\ell^n[3]) = 0$ for every n.

The section $Z_0 = \{h = 0\} \subset Z$ has equation $x^2 + zw = 0$, an A_1 surface singularity. Its induced resolution $\pi_0 : T_0 \to Z_0$ has $T_0 \simeq \mathrm{Tot}(\mathcal{O}_{\mathbf{P}^1}(-2))$: it is the kernel bundle of the surjection $\mathcal{O}(-1)^{\oplus 2} \xrightarrow{(s,t)} \mathcal{O}$. The exceptional curve is the zero section and has self-intersection -2. With compatible generators, the punctual intersection form is multiplication by -2, which is invertible over every $\mathbf{Q}_\ell$, including $\mathbf{Q}_2$. Write $\mathbf{Q}_{\ell,\{0\}}$ for the constant sheaf on the vertex, extended by zero to Z_0. For the inclusion $i : Z_0 \hookrightarrow Z$, the boundary criterion and proper base change therefore give

$$i^*\,\mathrm{IC}_Z[-1] \simeq R\pi_{0*}\mathbf{Q}_\ell[2] \simeq \mathrm{IC}_{Z_0} \oplus \mathbf{Q}_{\ell,\{0\}}(-1).$$

These algebraic identities remain valid on qcqs analytic neighborhoods. Thus shifted restriction preserves perversity and introduces a punctual summand.

For the vertex inclusion $v : \{0\} \hookrightarrow Z$, proper base change identifies the boundary complex with $v_*v^*R\pi_*\mathbf{Q}_\ell[3] \simeq v_*R\Gamma(\mathbf{P}^1, \mathbf{Q}_\ell)[3]$. It is supported entirely in the zero fiber of h, so its nearby cycles vanish and its vanishing cycles are nonzero. The second vanishing statement in (2.4) therefore fails. This boundary calculation explains the role of the second condition in Proposition 2.6.

5.2. **A projective family with nonsemisimple monodromy.**

Theorem 5.3. *Let π be a uniformizer of K, and consider the closed annulus $A = \mathrm{Spa}(K\langle T/\pi, \pi^2/T\rangle)$, where $|\pi|^2 \le |T| \le |\pi|$. There exists a smooth projective elliptic curve $f : \mathcal{E} \to A$ such that $R^1 f_{C*}\mathbf{Q}_\ell$ is nonsemisimple on A_C for every prime ℓ. More precisely, its rational Tate module $V_\ell(\mathcal{E}_C)$ fits into a nonsplit exact sequence*

$$0 \longrightarrow \mathbf{Q}_\ell(1) \longrightarrow V_\ell(\mathcal{E}_C) \longrightarrow \mathbf{Q}_\ell \longrightarrow 0, \tag{5.1}$$

whose extension class is the Kummer class of T. The complex $Rf_*\mathbf{Q}_\ell$ has the decomposition of Corollary 5.1.

Proof. Step 1. Construct the elliptic family. The universal Tate Weierstrass equation has coefficients given by integral power series and discriminant $\Delta(q) = q\prod_{n\ge1}(1-q^n)^{24}$. Substitute $q = T$. Since $\|T\| \le |\pi| < 1$, the coefficients converge in $R = K\langle T/\pi, \pi^2/T\rangle$. The element T is a unit, and the product converges to a unit because its factors are units converging uniformly to 1. Thus $\Delta(T)$ is a unit. The equation defines a smooth cubic in $\mathbf{P}^2_R$ with its usual section, whose relative analytification is the required smooth projective elliptic curve.

Step 2. Identify the torsion extension. For every positive integer N, the torsion of the Tate curve has the presentation

$$\mathcal{E}[N] = \coprod_{a=0}^{N-1} \big(\mathrm{Spec}\, R[z]/(z^N - T^a)\big)^{\mathrm{an}}, \qquad 0 \to \mu_N \to \mathcal{E}[N] \to \mathbf{Z}/N \to 0. \tag{5.2}$$

Here analytification is relative over $\mathrm{Spa}(R)$. The components carry the Tate group law, and the component index gives the quotient map. The inverse image of $1 \in \mathbf{Z}/N$ is the torsor $z^N = T$, so the extension class is the Kummer class of T; see [Hua17, §2], following [KM85, §8.7]. Taking $N = \ell^n$, passing to the inverse limit, and tensoring with $\mathbf{Q}_\ell$ gives (5.1).

Step 3. Prove nonsplitting over C. An analytic unit on A_C has an integer winding number $w : \mathcal{O}(A_C)^\times \to \mathbf{Z}$, with $w(uv) = w(u)+w(v)$ and $w(T) = 1$. It is the slope of the logarithmic Gauss norm along the annular skeleton. A unit has no zeros on the annulus, so its Laurent Newton polygon has a constant dominant exponent, and multiplication adds these exponents. Equivalently, one may write $u = aT^{w(u)}(1+h)$, with $a \in C^\times$, $h \in \mathcal{O}(A_C)$, and $\|h\| < 1$; see [Ram05, p. 213, (2.4.6)].

The Kummer sequence injects $\mathcal{O}(A_C)^\times/(\mathcal{O}(A_C)^\times)^{\ell^n}$ into $H^1(A_C, \mu_{\ell^n})$. Let c_n be the Kummer class of T, and let $c \in H^1(A_C, \mathbf{Z}_\ell(1))$ be the compatible integral class. The comparison with the inverse limit has no $\varprojlim^1 H^0$ term, since the transition maps $H^0(A_C, \mu_{\ell^{n+1}}) \to H^0(A_C, \mu_{\ell^n})$ are surjective. If $\ell^m c = 0$, then for every n there would be a unit u_n with $T^{\ell^m} = u_n^{\ell^n}$. Winding numbers would give $\ell^m = \ell^n w(u_n)$, which is impossible for $n > m$. Thus c is nontorsion and remains nonzero after rationalization.

Both rank-one terms of (5.1) are constant on A_C, so their nonsplit extension is not semisimple. Dualizing and using $R^1 f_{C*}\mathbf{Q}_\ell \simeq V_\ell(\mathcal{E}_C)^\vee$ gives the assertion about $R^1 f_{C*}\mathbf{Q}_\ell$. Corollary 5.1 gives

$$Rf_*\mathbf{Q}_\ell \simeq \mathbf{Q}_\ell \oplus R^1 f_*\mathbf{Q}_\ell[-1] \oplus \mathbf{Q}_\ell(-1)[-2].$$

□

Remark 5.4. For a single Tate elliptic curve E_q/K with parameter $q \in K$, $0 < |q| < 1$, the same construction gives a nonsplit G_K-extension $0 \to \mathbf{Q}_\ell(1) \to V_\ell(E_q) \to \mathbf{Q}_\ell \to 0$. The nonzero valuation of q detects its rational Kummer class. In the annular family, nonsemisimplicity persists after base change to C.

5.3. **An isotropic upper bound for singular support.** Let k be a complete algebraically closed, non-trivially valued field of characteristic zero and residue characteristic p, and let $\Lambda = \mathbf{F}_{\ell^r}$, where $\ell \ne p$ is prime and $r \ge 1$. For a smooth rigid space M/k and $F \in D^b_{\mathrm{zc}}(M, \Lambda)$, write $SS(F) \subset T^*M$ for its singular support, defined by local test functions in [Zho25b, §8]. We first construct an analytic isotropic upper bound, then determine the singular support of sheaves supported in dimension at most one.

Proposition 5.5. *Let M/k be smooth affinoid of pure dimension n, and let $F \in D^b_{\mathrm{zc}}(M, \Lambda)$. There is a closed conic analytic subset $C_F \subset T^*M$ containing $SS(F)$ such that every irreducible component of C_F is isotropic on its smooth locus. In particular, every component has dimension at most n.*

Proof. Step 1. Reduce to proper direct images from smooth spaces. By [BH22, Proposition 3.6], the constructible category on M is thickly generated by finite pushforwards of constant sheaves with finite coefficients. Let $f : Y \to M$ be such a finite morphism. Passing to the reduction and using the Mayer–Vietoris triangles for closed covers, we reduce to an irreducible Y. Choose a resolution $\pi : \widetilde{Y} \to Y$ that is an isomorphism outside

a closed subset Z, and put $E = \pi^{-1}Z$. For the inclusion $i : Z \hookrightarrow Y$, proper base change gives the abstract blow-up triangle

$$\Lambda_Y \longrightarrow R\pi_*\Lambda_{\widetilde{Y}} \oplus i_*\Lambda_Z \longrightarrow R(\pi|_E)_*\Lambda_E \longrightarrow \Lambda_Y[1].$$

The dimensions of Z and E are smaller than that of Y. Repeating the construction terminates, so F lies in the thick subcategory generated by finitely many $Rg_{a*}\Lambda_{V_a}$, with $g_a : V_a \to M$ proper and V_a smooth.

Step 2. Construct the critical incidence set. For a proper map $g : V \to M$ with V smooth, let $dg^* : g^*T^*M \to T^*V$ be the cotangent map and set

$$Z_g = \{(v, \xi) \in V \times_M T^*M : dg_v^*(\xi) = 0\}, \qquad C_g = \mathrm{im}(Z_g \to T^*M).$$

The equations define the zero locus of a section of the pullback of T^*V, so Z_g is a closed analytic subspace. The projection $V \times_M T^*M \to T^*M$ is proper, and its restriction to Z_g is proper. The proper mapping theorem gives C_g a reduced closed analytic structure. It is conic because the equations are homogeneous in ξ. A constant sheaf on a smooth space has singular support equal to the zero section. The inclusion under proper pushforward [Zho25b, Lemma 8.8(2)] therefore gives $SS(Rg_*\Lambda_V) \subseteq C_g$.

Step 3. Prove isotropy. Let λ_M be the tautological one-form on T^*M. Choose local coordinates x_i on M, write $g_i = x_i \circ g$, and let ξ_i be the corresponding cotangent coordinates. The pullback of λ_M to Z_g is $\sum_i \xi_i\, dg_i$, which vanishes by the defining equations of Z_g. Work over an affinoid open in T^*M, and let C be an irreducible component of C_g there. Choose an irreducible component of $(Z_g)_{\mathrm{red}}$ whose image is C. Such a component exists because there are finitely many irreducible components of the source over the chosen affinoid, their proper images are closed, and their union covers C. Resolving this component gives a proper surjection $h : \widetilde{Z} \to C$, with $\widetilde{Z}$ smooth and $h^*\lambda_M = 0$. Generic smoothness for proper morphisms [BH22, Theorem 2.29] gives a dense smooth open $C^\circ \subset C^{\mathrm{sm}}$ over which h is smooth and surjective. The restriction $h^\circ : h^{-1}(C^\circ) \to C^\circ$ is faithfully flat, and the cotangent sequence gives an injection $(h^\circ)^*\Omega^1_{C^\circ/k} \hookrightarrow \Omega^1_{h^{-1}(C^\circ)/k}$. Thus $h^*\lambda_M = 0$ implies $(h^\circ)^*(\lambda_M|_{C^\circ}) = 0$, and faithful flatness gives $\lambda_M|_{C^\circ} = 0$. By density, this form and its exterior derivative vanish on C^{sm}. Hence C is isotropic and has dimension at most n. The argument is local on T^*M.

Step 4. Assemble the upper bound. For the finitely many maps chosen in Step 1, put $C_F = \bigcup_a C_{g_a}$. This is a closed conic analytic subset whose irreducible components are isotropic. The definition of singular support using test functions and exactness of vanishing cycles show that containment in a fixed closed conic set is stable under shifts, cones, and direct summands. Hence $SS(F) \subseteq C_F$. □

Lemma 5.6. *Let D be a smooth disc over k with coordinate t, let $j : D^\times = D \setminus \{0\} \hookrightarrow D$, and let L be a finite-rank Λ-local system on $D^\times$, trivialized by a finite Kummer cover $t = s^N$ for some $N \geq 1$. Let V be a geometric fiber of L, and let $T \in \mathrm{GL}_\Lambda(V)$ be the action of a generator of its Kummer monodromy group. Put $V^T = \ker(T - \mathrm{id}_V)$ and $Q = j_{!*}(L[1])$. If h is an analytic function on D with $h(0) = 0$ and order $m \geq 1$ at 0, then, after identifying the nearby fibers with V, there is an exact sequence of Λ-vector spaces*

$$0 \longrightarrow V^T \longrightarrow V^{\oplus m} \longrightarrow \mathcal{H}^0\big(\phi_h(Q)[-1]\big)_0 \longrightarrow 0.$$

In particular, $\dim_\Lambda \mathcal{H}^0(\phi_h(Q)[-1])_0 = m \dim_\Lambda V - \dim_\Lambda V^T$.

Proof. Write $h(t) = at^m u(t)$, where $a \in k^\times$ and $u(0) = 1$. Since k has characteristic zero, the analytic implicit function theorem at $z = 1$ gives an analytic function $z(t)$ satisfying $z(t)^m = u(t)$ and $z(0) = 1$ after shrinking D. Choose an m-th root of a in k. The coordinate $a^{1/m}tz(t)$ then reduces h to t^m. This argument applies to every $m \geq 1$; when $p \mid m$, the radius of D may be decreased further.

The stalk of the intermediate extension at 0 is $Q_0 = V^T[1]$, and the geometric Milnor fiber of t^m consists of m points. The stalk of the nearby cycles is therefore $(V^{\oplus m})[1]$ by [Zho25b, Proposition 2.7]. Choose compatible fiber identifications on a Kummer cover trivializing L. An invariant section $v \in V^T$ specializes to its transported values at the m points. Projection to any one factor is therefore the inclusion $V^T \hookrightarrow V$, so the specialization map is injective. The triangle defining vanishing cycles gives the stated exact sequence and dimension formula. □

For a smooth locally closed subspace $S \subset M$, let $\overline{T^*_S M}$ be the closure of its conormal bundle in T^*M.

Proposition 5.7. *Let M/k be smooth, quasi-compact, and pure-dimensional of dimension n. If $F \in D^b_{\mathrm{zc}}(M, \Lambda)$ has support of dimension at most one, then $SS(F)$ is an analytic Lagrangian subset of T^*M. More precisely, it is a finite union of conormal closures of smooth curves and cotangent fibers at classical points.*

Proof. Step 1. Reduce to a simple perverse sheaf. By [Zho25b, Proposition 8.11(2)], $SS(F)$ is the union of the singular supports of its finitely many simple perverse constituents. It suffices to treat one such constituent P. If P is supported at a point x, then $SS(P) = T^*_x M$ by [Zho25b, Lemma 8.8(2)]. Otherwise, let C be its support, viewed as a reduced irreducible curve, and let $g : \widetilde{C} \to C \hookrightarrow M$ be its finite normalization. The curve $\widetilde{C}$ is smooth because k has characteristic zero. Finite pushforward is perverse t-exact and commutes with intermediate extension [BH22, Theorem 4.2(5)–(7)]. Thus there are a dense open immersion $\widetilde{j} : W \hookrightarrow \widetilde{C}$ and a nonzero lisse Λ-local system L on W such that, with $Q = \widetilde{j}_{!*}(L[1])$, we have $P = g_* Q$. Let S be the finite set where $Q[-1]$ is not lisse, and put $R = \{y \in \widetilde{C} : dg_y = 0\}$, which is finite because g is generically an immersion. We will prove

$$SS(P) = \overline{T^*_{C^{\mathrm{sm}}} M} \cup \bigcup_{x \in g(S \cup R)} T^*_x M. \tag{5.3}$$

Step 2. Bound the singular support. On the smooth lisse locus of $\widetilde{C}$, the singular support of Q is the zero section. The inclusion for singular support under proper pushforward [Zho25b, Lemma 8.8(2) and Example 8.14(1)] therefore places $SS(P)$ in the right-hand side of (5.3). On the smooth lisse locus of C, the calculation for a closed immersion gives the conormal bundle inside $SS(P)$. Taking its closure gives $\overline{T^*_{C^{\mathrm{sm}}} M} \subseteq SS(P)$. It remains to detect the indicated cotangent fibers.

Step 3. Compute the local vanishing cycles. Fix a classical point $x \in C(k)$ and a local function f on M with $f(x) = 0$. Suppose that $h_y = f \circ g$ is nonconstant of order $m_y \geq 1$ at each $y \in g^{-1}(x)$. Since Λ is finite, the restriction of L to a punctured disc around y is trivialized by a finite étale frame cover. After shrinking the disc, this cover becomes a disjoint union of Kummer covers by [Zho25b, proof of Proposition 8.12 and footnote 11]. Thus L satisfies the hypothesis on the Kummer cover in Lemma 5.6. Write V_y for its fiber, $r_y = \dim_\Lambda V_y$, and T_y for its local monodromy. Applying Lemma 5.6 to each branch of the normalization and using compatibility of vanishing cycles with finite proper pushforward gives

$$\dim_\Lambda \mathcal{H}^0\big(\phi_f(P)[-1]\big)_x = \sum_{y \in g^{-1}(x)} \big(m_y r_y - \dim_\Lambda V_y^{T_y}\big). \tag{5.4}$$

Every summand is nonnegative.

Step 4. Detect the cotangent fibers. Suppose $x \in g(S \cup R)$, and fix $\xi \in T^*_x M(k)$. Choose a local function f_0 with $f_0(x) = 0$ and $df_0(x) = \xi$, and a linear combination z of local coordinates vanishing at x that is nonconstant on every branch of C above x. For all but finitely many $a \in k$, the function $f = f_0 + az^2$ is nonconstant on every branch and still satisfies $f(x) = 0$ and $df(x) = \xi$. If $y \in R$, then $m_y \geq 2$, and its summand in (5.4) is at least $r_y > 0$. If $y \in S$, the local monodromy is nontrivial, so $\dim_\Lambda V_y^{T_y} < r_y$, again giving a positive summand. At least one such y occurs, hence $\phi_f(P)_x \neq 0$. The definition using test functions gives $(x, \xi) \in SS(P)$. Since ξ was arbitrary, density of classical covectors and closedness give $T^*_x M \subseteq SS(P)$. Together with Step 2, this proves (5.3).

Step 5. Prove analyticity and the Lagrangian property. Put $J = \mathrm{im}(g^*\Omega^1_M \to \Omega^1_{\widetilde{C}})$ and $E = \ker(g^*\Omega^1_M \to J)$. The sheaf J is torsion-free of rank one on the smooth curve $\widetilde{C}$, hence invertible. Thus $0 \to E \to g^*\Omega^1_M \to J \to 0$ is locally split, and E is a vector bundle of rank $n-1$. Its total space is a closed subspace of $\widetilde{C} \times_M T^*M$, whose projection to T^*M is finite because g is finite. Over the smooth locus of C, it identifies with the conormal bundle. Its reduced image is therefore the conormal closure, and the map is finite and birational onto that image. Since $\mathrm{Tot}(E)$ is smooth and normal, it is the normalization of this closure, which is closed analytic and pure of dimension n.

The tautological one-form pulls back to zero on $\mathrm{Tot}(E)$. The normalization is an isomorphism over the regular locus of the conormal closure, so the canonical symplectic form vanishes there. The closure is thus Lagrangian, as is each cotangent fiber $T^*_x M$. Formula (5.3) expresses $SS(P)$ as a finite union of analytic Lagrangian components. Taking the finite union over the perverse constituents of F proves the proposition. □

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